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Biomechanics-informed Non-rigid Medical Image Registration with Elasticity Theories

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***Abstract*— Biomechanical modelling of soft tissue provides a method for constraining medical image registration, such that the estimated spatial transformation is considered biophysically plausible. Existing methods either directly optimize the loss function containing the biomechanical-constrained regularization term over deformations, which takes much computational time, or are trained using biomechanically plausible data generated via finite element simulation, which is cumbersome. This work first instantiates the recently-proposed physics-informed neural networks (PINNs) to 3D elastic models that are used to establish the partial differential equations (PDEs) representing physics laws of biomechanical constraints to be satisfied. The registration algorithm that aligns point sets considering PINN-imposed biomechanics (i.e., the forward problem) is then formulated. In addition, the inverse problem and its algorithm of physical parameter (i.e., material property) estimation along with the registration are also formulated and developed. We carefully compare linear and nonlinear elasticity theories' capabilities in solving both tasks of forward registration and inverse physical parameter identification under PINNs respectively. Furthermore, two specific network configurations that leverage one common branch or two individual branches to predict deformation vectors and biomechanical states are also constructed and compared. The proposed PINNs-based registration approaches have been extensively evaluated with three experiments, that is single and multiple patient MRI-US registration using clinical MRI-US pairs, and registration using pairs of undeformed MR images from clinical cases of prostate cancer biopsy and deformed counterparts with finite-element-computed ground-truth deformation. Results demonstrate that the proposed methods achieve state-of-the-art performances compared to biomechanical-model-based and learning-based registration approaches, and the biomechanical constraints of soft tissues have been successfully warranted after registration. The codes are available at `https://github.com/ZheMin-1992/Registration_PINNs`.**

***Index Terms*— Medical Image Registration, Biomechanical Constraints, Physics-informed Neural Network.**

I. Introduction

MULTI-MODAL medical image registration enables access to clinically important information from different imaging modalities by spatially aligning them, for tasks such as surgical and interventional guidance [1], [2]. Perhaps due to the complementary nature between cross-modality images, designing a robust objective function or an unsupervised loss function is in general highly challenging, for classical or learning-based algorithms, respectively. This work investigates an example of such cross-modality registration, for establishing spatial correspondence between preoperative MR and 3D intraoperative transrectal ultrasound (TRUS) images from the same patients. Indeed, most previously proposed approaches utilised correspondent features from both images, for either iterative optimisation algorithms [3] or neural network training [4]. The inevitable sparsity of these available anatomical features, such as the boundaries of prostate gland and other zonal structures, necessitates the addition of transformation smoothness constraints. Hu et al [4] illustrated examples showing that, without imposing smoothness constraints on the registration-estimated transformation, highly distorted local deformation occurred which led to poorer target registration errors (TREs) in these areas. In addition to heuristically designed deformation regularisation, such as L^2 norm of local displacement and bending energy, displacement constraints originated from solid mechanics [3], [5], have also demonstrated benefits in this application, with an arguably flexible and purposive approach through its soft tissue modelling physics [6].

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A. Related Work

1) The Forward Registration Problem of Deformation Estimation: Different from voxelised volumetric images with rectangular grids, point sets are in general unstructured and unordered [7] for efficiently yet sparsely representing geometries or shapes. The classical point set registration method CPD estimates the optimal deformations by maximizing the likelihood that the target point set is drawn from the deformed Gaussian Mixture Models (GMMs) representing the source point set, together with minimizing the regularizer spatial smoothness with the assumption that nearby points are considered to move "coherently" as a group [8]. In contrast, GMM-FEM minimizes a different regularizer of biomechanical priors implicitly represented by finite-element model (FEM) [9].

PointNet was proposed to represent such point sets [7]. Originally designed for classification and segmentation tasks, PointNet was also adopted for learning-based rigid registration that either *1)* first establishes point correspondences in the feature spaces, with which then estimates the rigid transformation using closed-form solutions such as singular value decomposition [10], or *2)* directly aligns with learned feature representations to regress the rigid transformation parameters [11]. Among non-rigid registration approaches, Free Point Transformer (FPT) [2], [12] is an example that utilises PointNet to extract features to predict source-point-wise displacement vectors, trained with the composition of Chamfer loss [13] and/or negative log-likelihood function of Gaussian Mixture Models [2].

In [14], an adapted PointNet [7] was proposed using (FEM)-simulated training data to predict nodal displacement vectors for prostate meshes with unseen patients. In [15], FEM was first proposed to generate displacements for source point sets with boundary conditions established from an independent non-rigid iterative closest point (ICP) [16] procedure between prostate surfaces, before a network was trained using the FEM-generated transformations [15]. Biomechanical constraints have also been investigated in motion modelling and deformable registration, for other organs, such as liver [17], brain [18] and heart [19].

In the field of medical image registration, for both categories of classical and data-driven registration approaches, there exists work that utilizes biomechanical constraints to regularize the predicted deformation [20]. Burger et al. [21] presents a robust numerical scheme of image registration with hyperealsticity regularizer, with the aim of modelling large nonlinear deformations while preserving diffeomorphism. Genet et al. [22] presents the novel finite strain formulation of the equilibrium gap regularization in the context of finite-element-based image registration, demonstrating greater effectiveness and robustness compared with the strain-energy-based regularization. More recently, Genet et al. extends the equilibrium gap principle to the large deformation setting, and also presents a novel regularization term to control surface tractions [23]. WarpPINNs [24] minimizes a combination of the intensity difference between the deformed and target images, and the regularization term formulated using the neo-Hookean hyperelastic function. The voxel-wise displacement field is predicted using a fully connected network, while the near-incompressibility of cardiac tissue is considered by penalizing the Jacobian determinants of the cardiac voxels' displacements more heavily than those of the background tissues. Alvarez et al. [25] promotes the use of equilibrium gap principle against the bending energy in the regularization of registration, and has derived its local analytical voxel-wise form, which necessitates computing first and second-order derivatives of the displacement field.

2) Linear and Nonlinear Elasticities: Linear elasticity models assuming a linear relationship between stress and strain, are only effective for modelling small deformations under low-stress conditions [26]. Nonlinear elasticity models built on more complex constitutive models and strain energy functions, are better suited for capturing large deformations and nonlinear material behavior [26]. For example, the anisotropic viscoelasticity constitutive models [27] were utilised for simulating soft tissues' deformations, led to more realistic organ (e.g., liver) geometries than linear models with desired properties such as stress dissipation [28]. The nonlinear finite-element-analysis was developed for modelling-fidelity surgical simulations [29]. However, the choices of hyperviscoelastic, hyperelastic, and linear elastic constitutive models were demonstrated to be not important in estimating brain shifts for an image-guided neurosurgery procedure [30]. *It is unclear whether a more complex nonlinear model (i.e., both geometrical and constitutive models) could make a difference in the non-rigid point set registration problem.*

3) The Inverse Problem of Material Property Estimation: Material property estimation aims to quantify mechanical properties of materials (e.g., soft tissues), which can be used to develop customized surgical plans that could improve surgical outcomes [1], [31]. Perhaps more interestingly, it may assist disease detection and localisation, as studies frequently found *in vitro* tumour has distinct mechanical properties compared to the healthy tissue [32]. The identifications of bulk and shear modulus, under both linear elasticity and hyperelasticity models, were explored using specimens with simple one-or-two layers structures through experimental tests (e.g., uniaxial tension or compression) [33]. The authors revealed multiple findings about under what conditions parameters are identifiable, for example, to compute material parameters of a single-layered specimen, a uniaxial tensile test either together with lateral strain measurement, with torsion or with a shear experiment is needed [33]. We note that circumstances are much more complicated in both soft-tissue deformation modelling where exact values of stress and strain are unknown, and image registration where displacement vectors of voxels need to be estimated. *It is thus unknown whether material properties are identifiable or not, within the challenging scenario of image registration where boundary conditions also need to be estimated.*

B. The Proposed Method

This work investigates an alternative approach to encode biomechanical constraints represented by a system of partial

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differential equations (PDEs), which is solved simultaneously with minimising a registration loss. With the exemplary application of registering MR and TRUS prostate images, we *1)* propose an approach that represents prostate point displacements using PointNet, previously adopted in this application [2]; *2)* develop physics-informed neural network (PINNs) for imposing elastic constraints on the estimated displacements; *3)* formulate an end-to-end registration network training algorithm, by minimising surface distance as estimated boundary conditions in the PDEs; *4)* designate the inverse problem of estimating material properties (e.g., the ratios of Young's Modulus between different sub-regions) together with the forward registration problem; *5)* compare the linear and nonlinear elasticities' capabilities in solving the forward and inverse problems.

1) Differences from Existing Work: Our approach differs from those [21]–[23] in the following aspects. First, all the three methods [21]–[23] belong to the category of conventional registration or tracking algorithms in which the regularization term, defined with respect to the displacement vectors, is directly added to the objective function and thus the optimization speed is slow. In contrast, we use networks to predict point-wise displacement vectors and stress tensors, which implicitly encodes biomechanical constraints after being trained and thus exhibits a fast inference speed. Second, only the strain energy function is leveraged in the optimization in [21]–[23]. In contrast, our approach leverages several biomechanics-related terms, including those constructed from the static equilibrium equations, the constitutive equations and the strain energy term. Moreover, we present two versions of the registration algorithms using both linear and nonlinear elasticities respectively.

Compared with WarpPINNs [24] and [25], our work differs significantly in several important aspects. First, WarpPINNs and [25] only predicts the displacement vectors while our approach leverages two network branches to predict both displacement vectors and stress. Second, our approach works with the unordered point sets while WarpPINNs [24] and [25] are tailored specifically for medical image registration. Third, we also investigate the inverse problem where the material properties (i.e., the ratio of Young's Modulus between two sub-regions) are estimated along with the registration. Fourth, we have formulated both the forward and inverse problems using both the linear and nonlinear elasticity theories whose performances have been carefully compared. Last but not least, we leverage additional constitutive and static equilibrium equations to formulate the biomechanical loss terms, whose effectiveness has been validated in Sec. IV-F.

It is noted that our work takes unordered point sets while [21]–[25] deals with medical images. In this regard, our work differs evidently from biomechanics-constrained optimisation-based point set registration method GMM-FEM [9] and learning-based methods [15], in that the finite element modelling process being both troublesome and time consuming is not needed.

2) Contributions: The contributions are summarised as follows. **1)** We developed a patient-specific registration algorithm combining PointNet and PINNs, which aligns prostate glands segmented from MR and TRUS images, subject to biomechanical constraints exerted from soft-tissue-modelling PDEs (Fig. 1). **2)** We demonstrated that both the biomechanically-regularised deformation and the TRE-reducing correspondence can be generalised to unseen new patients, with the PINN-based registration network trained on a small number of training examples. **3)** We presented a set of experimental results for evaluating the theoretical and clinical efficacy in soft tissue modelling within registration algorithms, with statistical significance, using finite element (FE)-based ground-truth and independent landmark-based TREs, respectively. **4)** We demonstrated extensively that nonlinear elasticity shows no statistical significance against linear models in computing point-wise displacement vectors but their respective benefits may depend on specific patients, with FE computed ground-truth. **5)** We formulated and tackled the inverse parameter estimation problem, under the joint optimisation scheme of registration and parameter identification using PINNs, whose solutions can be accurately found by locating saddle points of the optimisation.

3) Organization of this Paper: In Section II, we describe the proposed registration algorithm with linear elasticity. In Section III, we present the counterpart algorithm with nonlinear elasticity, and the inverse problem. In Section IV, we present experiments and results. In Section V, we discuss the limitations of this work and future work. In Section VI, we conclude the paper.

II. Non-Rigid Point Set Registration using Linear Elasticity

Let $\mathbf{P}_{\mathcal{S}} \in \mathbb{R}^{N_s \times 3}$ and $\mathbf{P}_{\mathcal{T}} \in \mathbb{R}^{N_t \times 3}$ be a pair of source and target point sets with individual points being $\mathbf{p}_s = [p_s^x, p_s^y, p_s^z]^\mathsf{T} \in \mathbb{R}^3$ and $\mathbf{p}_t = [p_t^x, p_t^y, p_t^z]^\mathsf{T} \in \mathbb{R}^3$, where $N_s \in \mathbb{N}^+$ and $N_t \in \mathbb{N}^+$ are number of points, $s \in \{1, ..., N_s\}$ and $t \in \{1, ..., N_t\}$ are indexes of points. The non-rigid point set registration problem is to find point-wise displacement vectors $\mathbf{D}_{\mathcal{S}} \in \mathbb{R}^{N_s \times 3}$ with individual displacement being $\mathbf{d}_s = [d_s^x, d_s^y, d_s^z]^\mathsf{T} \in \mathbb{R}^3$, with which the warped source point set $\mathsf{T}(\mathbf{P}_{\mathcal{S}}) = \mathbf{P}_{\mathcal{S}} + \mathbf{D}_{\mathcal{S}}$ aligns with $\mathbf{P}_{\mathcal{T}}$. For clarity, we also define the follows for each s,

$$\mathsf{T}(\mathbf{p}_s) = \mathbf{p}_s + \mathbf{d}_s. \tag{1}$$

We additionally adopt notations $\mathbf{P}_{\mathcal{S}}^{\text{internal}}$ and $\mathbf{P}_{\mathcal{S}}^{\text{surface}}$, $\mathbf{P}_{\mathcal{T}}^{\text{internal}}$ and $\mathbf{P}_{\mathcal{T}}^{\text{surface}}$ to distinguish internal and surface points in $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$, respectively.

A. Physics-informed Neural Network (PINNs) for Non-rigid Registration with Biomechanical Constraints

Considering the physical laws that need to be satisfied, the non-rigid point set registration estimating displacement vectors $\mathbf{D}_{\mathcal{S}}$ is considered as the problem of seeking data-driven solutions to partial differential equations (PDEs). Let $k \in \mathbb{N}^+$ be the patient index and $\mathcal{P}_k$ be the set containing $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$ for the k-th patient, then the entire network denoted as $e_\theta(\mathcal{P}_k)$ with trainable parameters θ, consists of two sub-networks $g_{\theta_g}(\mathcal{P}_k)$ and $h_{\theta_h}(\mathcal{P}_k)$ as shown in Fig. 1, with completing parameter sets θ_g and θ_h, predicting displacement

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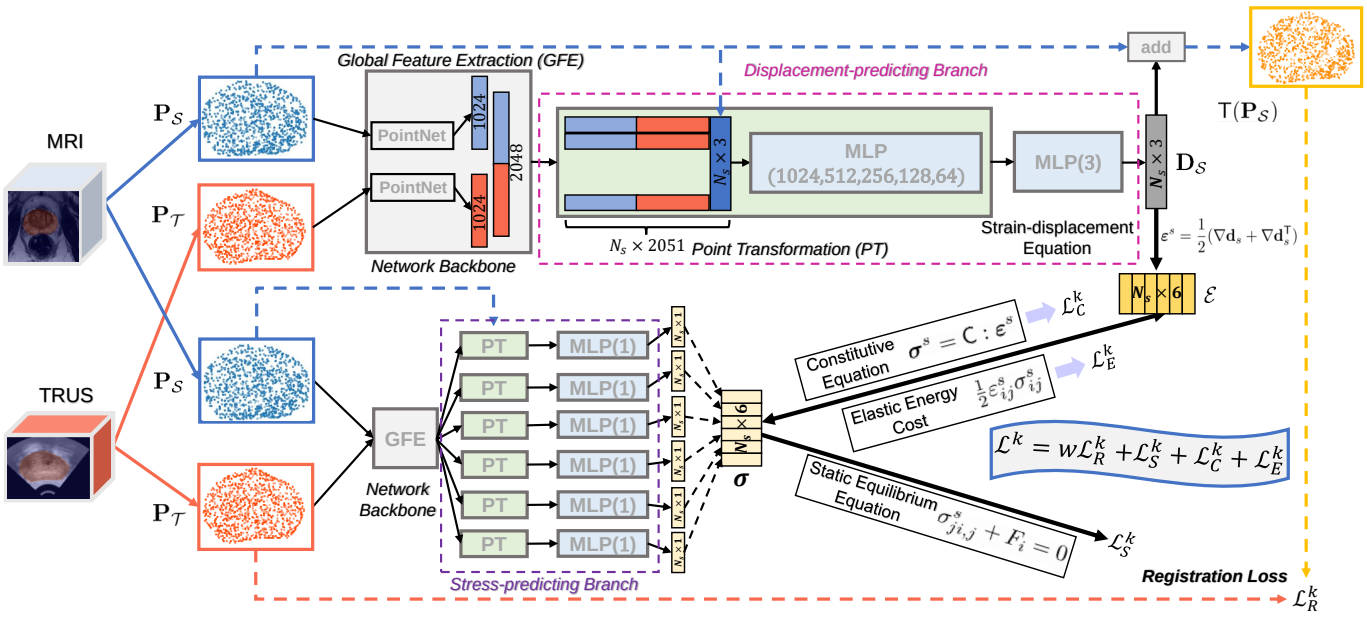


Fig. 1. The proposed non-rigid medical image registration framework using physics-informed neural networks (PINNs), whose inputs are a pair of source and target point sets $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$ extracted from MRI and TRUS volumes of the same patient, respectively. PINNs consist of $g_{\theta_g}(\mathcal{P}_k)$ predicting displacement vectors $\mathbf{D}_{\mathcal{S}}$ from which the point-wise strain tensors $\boldsymbol{\mathcal{E}}$ are further computed with the strain-displacement equation in Eq. (4), and $h_{\theta_h}(\mathcal{P}_k)$ predicting stress tensors $\boldsymbol{\sigma}$. The source point set $\mathbf{P}_{\mathcal{S}}$ added by $\mathbf{D}_{\mathcal{S}}$ results in the warped source point set $\mathbf{T}(\mathbf{P}_{\mathcal{S}})$, between which and $\mathbf{P}_{\mathcal{T}}$ the Chamfer loss $\mathcal{L}_R^k$ is computed using Eq. (7). $\mathcal{L}_S^k$ in Eq. (8) and $\mathcal{L}_C^k$ in Eq. (10) penalise deviations from the equality of the static equilibrium equation in Eq. (5) about $\boldsymbol{\sigma}$, and that of the constitutive equation in Eq. (6) about $\boldsymbol{\sigma}$ and $\boldsymbol{\mathcal{E}}$, respectively. $\mathcal{L}_E^k$ in Eq. (12) is the elastic energy cost that shall also be minimised.

vectors $\mathbf{D}_{\mathcal{S}}$ and stress tensors $\sigma \in \mathbb{R}^{N_s \times 6}$, respectively. With the capability of universal function approximation, physics-informed neural networks (PINNs) can be utilised to model physical laws(i.e., biomechanical constraints) represented by nonlinear PDEs [34], as will later be introduced in Section II-B. Let a function $f(\mathbf{p}_s, \mathbf{d}_s, \boldsymbol{\sigma}^s)$ of the source point $\mathbf{p}_s$, the displacement vector $\mathbf{d}_s$ and stress tensor $\boldsymbol{\sigma}^s$, be a PINN defining biomechanical constraints partially characterised by known material properties b_s:

$$f := f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) + f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, b_s) + f_{\text{energy}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s), \quad (2)$$

where x, y and z are spatial coordinates of $\mathbf{p}_s(x, y$ and z are actually equal to p_s^x, p_s^y, p_s^z, for brevity hereafter we use x, y and $z)$, the determination of b_s is detailedly described in Sect. IV, $f_{\text{static}}(\cdot)$, $f_{\text{const}}(\cdot)$ and $f_{\text{energy}}(\cdot)$ represent norms of residuals deviating from static equilibrium, constitutive equality and null elastic energy, respectively, as defined in the remainder Sect. II-B and Sect. II-C. The network parameters are optimised by minimising the follows

$$\mathcal{L}^k(\theta; \mathcal{P}_k) = \mathcal{L}_R^k(\theta_g; \mathcal{P}_k) + \mathcal{L}_F^k(\theta; \mathcal{P}_k), \quad (3)$$

where $\mathcal{L}_F^k(\theta; \mathcal{P}_k) = \sum_{s=1}^{N_s} f(\mathbf{p}_s, \mathbf{d}_s, \sigma^s)$ is the term concerning biomechanical constraints over all sampled source points, while $\mathcal{L}_R^k(\theta_g; \mathcal{P}_k)$ can be either (1) $\sum_{s=1}^{N_s} \left(||\mathbf{d}_s - \mathbf{d}_s^{\text{gt}}||_2^2\right)$ with $\mathbf{d}_s^{\text{gt}} \in \mathbb{R}^3$ denoting ground-truth displacement vectors of $\mathbf{p}_s$ under the supervised learning scheme (e.g., simulated data with known ground-truth deformations) and $||\bullet||_2$ denotes the L2 norm of a vector; or (2) $\phi(\mathsf{T}(\mathbf{P}_{\mathcal{S}}), \mathbf{P}_{\mathcal{T}})$ being the unsupervised loss (e.g., the Chamfer loss [13] for the purpose of aligning point sets) which measures goodness-of-prediction, resulting in a complete registration algorithm as described in Sect. II-C.

B. Governing Equations for Deforming Linear Elastic Organs adapted for Medical Image Registration

Linear elasticity is used as the first specific example to characterise prostate gland deformation between $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$, primarily due to contact with a moving ultrasound probe [1], [31]. There are three main governing equations, the strain-displacement equation describing the relationship between strain and displacement that should be satisfied (cf. Sect. II-B.1), the static equilibrium equation about the spatial derivatives of the stress (cf. Sect. II-B.2), the constitutive equation describing the relationship between the stress and strain that should be satisfied (cf. Sect. II-B.3). It is noted that both strain tensor $\boldsymbol{\varepsilon}^s$ and stress tensor $\boldsymbol{\sigma}^s$ are symmetric second-order tensors, i.e., $\varepsilon_{ij}^s = \varepsilon_{ji}^s$ and $\sigma_{ij}^s = \sigma_{ji}^s$ where $i \in \{x, y, z\}$ and $j \in \{x, y, z\}$. It is noted that the Voigt notations are leveraged to represent both strain and stress tensors (cf. Appendix VII-D).

1) Strain-displacement Equations: The strain-displacement equation (i.e., the kinematic equation) in tensor notation at a source point $\mathbf{p}_s$ is

$$\boldsymbol{\varepsilon}^s = \frac{1}{2}(\boldsymbol{\nabla}\mathbf{d}_s + \boldsymbol{\nabla}\mathbf{d}_s^{\mathsf{T}}), \quad (4)$$

where $\boldsymbol{\varepsilon}^s$ is the infinitesimal second-order Cauchy strain tensor at $\mathbf{p}_s$, ∇ denotes the nabla operator, $\nabla\mathbf{d}_s$ is the displacement gradient w.r.t. spatial coordinates x, y, z of $\mathbf{p}_s$.

In the registration process, displacement vectors $\mathbf{D}_{\mathcal{S}}$ are

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first predicted by $g_{\theta_g}(\mathcal{P}_k)$ from which Eq. (4) is then used to compute strain matrix $\boldsymbol{\mathcal{E}} \in \mathbb{R}^{N_s \times 6}$.

2) Static Equilibrium Equations: The spatial components of the Cauchy stress tensor $\boldsymbol{\sigma}^s$ at $\mathbf{p}_s$, predicted by $h_{\theta_h}(\mathcal{P}_k)$, satisfy the following equilibrium equation (i.e., equation of motion) in tensor notation

$$\boldsymbol{\nabla} \cdot \boldsymbol{\sigma}^s = \mathbf{0}. \tag{5}$$

In the registration process, the stress matrix $\boldsymbol{\sigma} \in \mathbb{R}^{N_s \times 6}$ is predicted by $h_{\theta_h}(\mathcal{P}_k)$, from which the spatial derivatives are computed and utilised in Eq. (5).

3) Constitutive Equations: The stress and strain tensors at $\mathbf{p}_s$ are related by the constitutive equation (i.e., the generalised Hooke's law) in tensor notation as

$$\boldsymbol{\sigma}^s = \mathsf{C}_s : \boldsymbol{\varepsilon}^s, \tag{6}$$

where C_s is the fourth-order elasticity tensor constructed using the material properties, the operator : here denotes the inner product of tensors where the summation over repeated indices is implied. The matrix form of C_s is presented in Eq. (37).

In the registration process, the stress matrix $\boldsymbol{\sigma}$ predicted by $h_{\theta_h}(\mathcal{P}_k)$ and strain matrix $\boldsymbol{\mathcal{E}}$ computed using Eq. (4) are utilised to construct Eq. (6).

For clarity, the explicit forms of the above three linear governing equations are presented in Eq. (30), Eq. (32) and Eq. (39), respectively.

C. A Non-Rigid Point Set Registration Algorithm using PINNs

Fig. 1 shows the schematic of the proposed non-rigid point set registration network, with the displacement-predicting branch $g_{\theta_g}(\mathcal{P}_k)$ and stress-predicting branch $h_{\theta_h}(\mathcal{P}_k)$. In what follows, we present the loss functions for single-pair patient-specific registration in Sect. II-C.1 and multi-patient generalisable learning algorithm in Sect. II-C.2 respectively.

1) Loss Functions for Single-Pair Patient-Specific Registration: The loss function includes four terms. First, the Chamfer loss $\phi(\mathsf{T}(\mathbf{P}_\mathcal{S}), \mathbf{P}_\mathcal{T})$ [13] is minimised to spatially align the two point sets as much as possible, and is given by

$$\begin{aligned}\mathcal{L}_R^k(\theta_g; \mathcal{P}_k) = &\frac{1}{|\widetilde{N}_t|}\Big(\sum_{t\in\widetilde{N}_t} \min_{s\in\widetilde{N}_s} ||\mathsf{T}(\mathbf{p}_s) - \mathbf{p}_t||_2^2\Big) \\ &+ \frac{1}{|\widetilde{N}_s|}\Big(\sum_{s\in\widetilde{N}_s} \min_{t\in\widetilde{N}_t} ||\mathsf{T}(\mathbf{p}_s) - \mathbf{p}_t||_2^2\Big),\end{aligned} \tag{7}$$

where $\widetilde{N}_s \subseteq \{1, ..., N_s\}$ and $\widetilde{N}_t \subseteq \{1, ..., N_t\}$ denote sets of utilised points being either the entire organ $\mathbf{P}_\mathcal{S}$ and $\mathbf{P}_\mathcal{T}$ or a subset region, e.g., surface points $\mathbf{P}_\mathcal{S}^{\text{surface}}$ and $\mathbf{P}_\mathcal{T}^{\text{surface}}$, $|\widetilde{N}_s|$ and $|\widetilde{N}_t|$ are numbers of utilised points, $\mathsf{T}(\mathbf{p}_s)$ is defined in Eq. (1). As can be seen from Eq. (7), what the Chamfer loss computes is the following: (a) for each point in the target point set $\mathbf{P}_\mathcal{T}$, it computes its distance (i.e., L2 norm) with the neareast point in the source point set $\mathsf{T}(\mathbf{P}_\mathcal{S})$, and then computes the average value of the sum of all the distances' squares over all target points; (b) repeat (a) with swapping the roles of $\mathbf{P}_\mathcal{S}$ and $\mathbf{P}_\mathcal{T}$; (c) returns the sum of both averages in (a) and (b) as the Chamfer loss.

Second, the deviation from the static equilibrium equation in Eq. (5) w.r.t. the stress $\boldsymbol{\sigma}^s$ is penalised by minimising $\mathcal{L}_S^k(\theta_h; \mathcal{P}_k)$ defined as

$$\mathcal{L}_S^k(\theta_h; \mathcal{P}_k) = \frac{1}{|\widetilde{N}_s|}\sum_{s=1}^{|\widetilde{N}_s|} f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s), \tag{8}$$

where the loss term with $\mathbf{p}_s$ in Einstein notation is

$$f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) = \sum_{i\in\{x,y,z\}} ||\sigma_{ji,j}^s||_2^2. \tag{9}$$

where $(\cdot)_{,j}$ is a shorthand for $\frac{\partial(\cdot)}{\partial(\mathbf{p}_s^j)}$.

Third, $\mathcal{L}_C^k(\theta; \mathcal{P}_k)$ regularises network-predicted $\boldsymbol{\sigma}$ and estimated strain $\boldsymbol{\mathcal{E}}$ to satisfy constitutive equations in Eq. (6), and is defined as

$$\mathcal{L}_C^k(\theta; \mathcal{P}_k) = \frac{1}{|\widetilde{N}_s|}\sum_{s=1}^{|\widetilde{N}_s|} f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, \mathsf{C}_s), \tag{10}$$

where the loss term with $\mathbf{p}_s$ in Einstein notation is

$$f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, \mathsf{C}_s) = \sum_{\langle i,j\rangle} ||\sigma_{ij}^s - \mathsf{C}_s^{ijkl}\varepsilon_{kl}^s||_2^2, \tag{11}$$

where $\langle i, j\rangle \in \{\langle x, x\rangle, \langle y, y\rangle, \langle z, z\rangle, \langle x, y\rangle, \langle x, z\rangle, \langle y, z\rangle\}$.

Fourth, the elastic energy cost $\mathcal{L}_E^k(\theta; \mathcal{P}_k)$ is also to be minimised

$$\mathcal{L}_E^k(\theta; \mathcal{P}_k) = \frac{1}{|\widetilde{N}_s|}\sum_{s}^{|\widetilde{N}_s|} f_{\text{energy}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s), \tag{12}$$

where the elastic energy term with $\mathbf{p}_s$ in Einstein notation is defined as

$$f_{\text{energy}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s) = \frac{1}{2}\varepsilon_{ij}^s\sigma_{ij}^s. \tag{13}$$

In the single-pair image registration for the given subject k, the overall training loss $\mathcal{L}^k(\theta; \mathcal{P}_k)$ is given by a ($w \in \mathbb{R}^+$)-weighted sum of these terms in Eq. (7), Eq. (8), Eq. (10) and Eq. (12)

$$w\mathcal{L}_R^k(\theta_g; \mathcal{P}_k) + \mathcal{L}_S^k(\theta_h, \mathcal{P}_k) + \mathcal{L}_C^k(\theta; \mathcal{P}_k) + \mathcal{L}_E^k(\theta; \mathcal{P}_k). \tag{14}$$

For clarity, the explicit forms of the above three biomechanical constraints related loss terms are presented in Eq. (42), Eq. (43) and Eq. (44) in Sec. VII-B, respectively .

2) Optimisation for a Multi-Patient Learning Algorithm: The above-described network can be adapted with minimal change in implementation, for a population-trained registration algorithm, by optimising network parameters θ with respect to an amortization loss:

$$\theta^\star = \arg\min_\theta \mathcal{L}(\theta; \mathcal{P}) = \arg\min_\theta E_k(\mathcal{L}^k(\theta; \mathcal{P}_k)), \tag{15}$$

where $\mathcal{P}$ is all the training data from multiple subjects, E_k denotes the expected value over all training examples and $\mathcal{L}^k(\theta; \mathcal{P}_k)$ is defined in Eq. (14).

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III. Non-rigid Medical Image Registration using Nonlinear Elasticity and the Inverse Problem

To account for large deformations, nonlinear elastic models are now investigated to derive the corresponding governing equations. In addition, two network configurations (i.e., Cfg1 and Cfg2) have been compared, where individual (cf. Fig. 1) and shared backbones (i.e., global feature extraction module that is based on PointNet [7]) are utilised for extracting features for two branches respectively. For clarity, the four learning models (combinations of two network configurations and two elasticity models (i.e., linear and nonlinear elasticity models)) are denoted as Linear Cfg1, Linear Cfg2, Nonlinear Cfg1 and Nonlinear Cfg2.

A. The Nonlinear Governing Equations and Loss Functions

In this Section, we present the nonlinear governing equations corresponding to those in Sect. II-B.

1) Nonlinear Strain-displacement Equations: The nonlinear strain-displacement equation at a source point $\mathbf{p}_s$ is

$$\boldsymbol{\varepsilon}^s = \frac{1}{2}(\nabla \mathbf{d}_s + \nabla \mathbf{d}_s^{\mathsf{T}} + \nabla \mathbf{d}_s^{\mathsf{T}} \nabla \mathbf{d}_s), \tag{16}$$

where $\boldsymbol{\varepsilon}^s$ is the *Green-Lagrangian strain* tensor at $\mathbf{p}_s$, $\nabla \mathbf{d}_s$ is the displacement gradient w.r.t. spatial coordinates x, y, z of $\mathbf{p}_s$. Comparing Eq. (16) and Eq. (4), we can notice that the second-order terms are retained in the nonlinear model. As shown in Fig. 1, Eq. (16) is utilised to derive point-wise strain ε^s from the predicted displacement vector $\mathbf{d}_s$.

2) Static Equilibrium Equations: In nonlinear elasticity, $\boldsymbol{\sigma}^s$ predicted by $h_{\theta_h}(\mathcal{P}_k)$ would be a *2nd Piola-Kirchhoff stress* tensor at $\mathbf{p}_s$, satisfying the similar equilibrium equation in Eq. (5). The corresponding loss function $\mathcal{L}_S^k(\theta_h; \mathcal{P}_k)$ is computed in a similar way as that in Eq. (8), and the involved function $f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s)$ is defined in the similar way as that in the linear case

$$f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) = \sum_{i \in \{x,y,z\}} ||\sigma^s_{ji,j}||_2^2, \tag{17}$$

where $(\cdot)_{,j}$ is a shorthand for $\frac{\partial(\cdot)}{\partial(p_s^j)}$ for the second Piola-Kirchhoff stress tensor and $(\cdot)_{,j}$ denotes $\frac{\partial(\cdot)}{\partial(p_s^j + d_s^j)}$ for the Cauchy stress tensor. The detailed explicit forms of $f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s)$ using the second Piola-Kirchhoff and Cauchy stress tensors are presented in Eq. (47) and Eq. (48), respectively.

3) Nonlinear Constitutive Equations and Loss Terms: The stress and displacement gradients are further constrained by the nonlinear constitutive equation as

$$\boldsymbol{\sigma}^s = \mu_s J_s^{-1}(\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}} - \mathbf{I}_{3\times 3}) + \lambda_s (J_s - 1)\mathbf{I}_{3\times 3}, \tag{18}$$

where here $\boldsymbol{\sigma}^s \in \mathbb{R}^3$ is reloaded as the matrix form of Cauchy stress tensor and is defined in Appendix VII-D, the deformation gradient $\mathbf{F}_s \in \mathbb{R}^{3\times 3}$ at $\mathbf{p}_s$ is defined as $\mathbf{F}_s = \mathbf{I}_{3\times 3} + \frac{\partial \mathbf{d}_s}{\partial \mathbf{p}_s}$, where $J_s = \det(\mathbf{F}_s) \in \mathbb{R}$ is the determinant of $\mathbf{F}_s$, $\lambda_s \in \mathbb{R}$ and $\mu_s \in \mathbb{R}$ are Lamé parameters at $\mathbf{p}_s$ which are computed with Young's Modulus E_s and Possion's ratio v_s using Eq. (40) and Eq. (41), respectively. The PDEs that comprise $\mathcal{L}_C^k(\theta; \mathcal{P}_k)$ in Eq. (14) and Eq. (21) are

$$\begin{aligned} &f_{\text{const}}(\mathbf{F}_s, \boldsymbol{\sigma}^s, \lambda_s, \mu_s) = f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, J_s, \lambda_s, \mu_s) \\ &= \sum_{i \in \{1,2,3\}} ||\sigma^s_{ii} - \mu_s J_s^{-1}((\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}})_{ii} - 1) + \lambda_s (J_s - 1)||_2^2 \\ &+ \sum_{\langle i,j \rangle \in \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 2,3\rangle\}} ||\sigma^s_{ij} - \mu_s J_s^{-1}(\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}})_{ij}||_2^2, \end{aligned} \tag{19}$$

where to maintain uniformity of notations here we define $\sigma^s_{11} = \sigma^s_{xx}$ (similar for index pairs $\langle 2,2\rangle$ and $\langle 3,3\rangle$) and $\sigma^s_{12} = \sigma^s_{xy}$ (similar for index pairs $\langle 1,3\rangle$ and $\langle 2,3\rangle$). It is noted that the first equality in Eq. (19) holds due to that the involved expression $\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}}$ can be represented using $\boldsymbol{\varepsilon}^s$, and J_s is omitted in the argument list of the function notation $f_{\text{const}}(\mathbf{F}_s, \boldsymbol{\sigma}^s, \lambda_s, \mu_s)$ since J_s is just a numerical value that can be directly computed using $\mathbf{F}_s$.

4) Nonlinear Elastic Energy Density Function: The elastic energy function $f_{\text{energy}}(\mathbf{F}_s, \lambda_s, \mu_s) = f_{\text{energy}}(\boldsymbol{\varepsilon}^s, J_s, \lambda_s, \mu_s)$ that forms $\mathcal{L}_E^k(\theta_h; \mathcal{P}_k)$ in Eq. (14) and Eq. (21) is

$$\begin{aligned} &f_{\text{energy}}(\mathbf{F}_s, \lambda_s, \mu_s) = f_{\text{energy}}(\boldsymbol{\varepsilon}^s, J_s, \lambda_s, \mu_s) \\ &= \frac{\mu_s}{2}\Big(\text{tr}(\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}}) - 3 - 2\ln J_s\Big) + \frac{\lambda_s}{2}\big(J_s - 1\big)^2\Big) \end{aligned} \tag{20}$$

where $\text{tr}(\bullet)$ is the trace of a matrix and $\text{tr}(\mathbf{F}_s \mathbf{F}_s^{\mathsf{T}})$ is represented by $\boldsymbol{\varepsilon}^s$.

B. The Non-rigid Point Set Registration with Nonlinear Elasticity

With the three involved functions $f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s)$, $f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, J_s, \lambda_s, \mu_s)$ and $f_{\text{energy}}(\boldsymbol{\varepsilon}^s, J_s, \lambda_s, \mu_s)$ defined in Eq. (17), Eq. (19) and Eq. (20) respectively, the corresponding loss functions $\mathcal{L}_S^k(\theta_h; \mathcal{P}_k)$, $\mathcal{L}_C^k(\theta; \mathcal{P}_k)$ and $\mathcal{L}_E^k(\theta; \mathcal{P}_k)$ in Eq. (8), Eq. (10) and Eq. (12) are computed. Together with $\mathcal{L}_R^k(\theta_g; \mathcal{P}_k)$ in Eq. (7), the overall training loss function in Eq. (14) can be constructed and re-computed for the nonlinear elasticity model.

C. The Inverse Problem

In Sect. II-C, we treat the registration as the forward problem using PINNs, which necessitates knowing material properties (e.g., Young's modulus and Possion's ratio). In contrast, the parameter (e.g., material properties) estimation is regarded as the inverse problem (i.e., the data-driven discovery of PDEs) under a similar PINNs framework.

As one typical example, let us consider two distinct compartments of the prostate gland, i.e., the peripheral zone (PZ) and transition zone (TZ), which exhibit different stiffness magnitudes [1], [31]. Let $\mathcal{P}_k^{\text{PZ}}$ and $\mathcal{P}_k^{\text{TZ}}$ denote points in PZ and TZ respectively, $E_k^{\text{PZ}} \in \mathbb{R}$ and $E_k^{\text{TZ}} \in \mathbb{R}$ be their respective Young's modulus values. The particular example of the inverse problem here, assuming that E_k^{TZ} is known, is to estimate the ratio $\beta_k \in \mathbb{R}$ of E_k^{PZ} to E_k^{TZ}. To this end, one additional learnable weight that functions as β_k is added into the network $e_\theta(\mathcal{P}_k)$. The loss function $\mathcal{L}_{\text{inverse}}^k(\theta, \beta_k; \mathcal{P}_k)$ for

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the inverse problem, under the joint learning scheme of point set registration and parameter estimation, is

$$\begin{aligned}\mathcal{L}^k_{\text{inverse}}(\theta,\beta_k;\mathcal{P}_k) = w\mathcal{L}^k_R(\theta_g;\mathcal{P}_k) + \mathcal{L}^k_S(\theta_h,\mathcal{P}_k) \\ + \mathcal{L}^k_C(\theta,\beta_k;\mathcal{P}_k) + \mathcal{L}^k_E(\theta_h,\beta_k;\mathcal{P}_k),\end{aligned} \tag{21}$$

where $\mathcal{L}^k_C(\theta,\beta_k;\mathcal{P}_k)$ and $\mathcal{L}^k_E(\theta,\beta_k;\mathcal{P}_k)$ can be expanded as

$$\mathcal{L}^k_{C/E}(\theta,\beta_k;\mathcal{P}_k) \equiv \mathcal{L}^k_{C/E}(\theta;\mathcal{P}^{\text{TZ}}_k, E^{\text{TZ}}_k) + \mathcal{L}^k_{C/E}(\theta;\mathcal{P}^{\text{PZ}}_k, \beta_k E^{\text{TZ}}_k), \tag{22}$$

where Lamé parameters λ_s and μ_s at $\mathbf{p}_s$ are computed using E^{TZ}_k and $\beta_k E^{\text{TZ}}_k$ depending on the sub-regions (i.e., $\mathcal{P}^{\text{PZ}}_k$ or $\mathcal{P}^{\text{TZ}}_k$) in which $\mathbf{p}_s$ falls.

The inverse problem of simultaneously predicting displacement vectors and estimating the ratios of Young's Mudulus $E^{\text{PZ}}_k/E^{\text{TZ}}_k$ by finding the optimal network parameters θ and β_k is now formally defined as

$$(\theta^\star,\beta^\star_k) = \arg\min_{(\theta,\beta_k)} \mathcal{L}^k_{\text{inverse}}(\theta,\beta_k;\mathcal{P}_k) \tag{23}$$

which is likely to be ill-posed, in that it contains unknowns in both registration-related and physics-laws-related loss terms (i.e., PDEs). Instead of seeking local minimums w.r.t material parameters which is likely to be unidentifiable and produces naive solutions, we are investigating saddle points which are much more interesting and possibly represent practically meaningful solutions, as will be demonstrated in Sect. IV-E.

IV. Experiments and Results

A. Datasets

The first dataset contained 77 pairs of MRI and TRUS volumetric images (both were resampled to $0.8 \times 0.8 \times 0.8$ mm^3) from the prostate cancer biopsy procedure, where the exemplar clinical application is to register pre-operative MRI images with TRUS images where prostate gland has been deformed due to surgical probe contact [31]. Each pair of point sets was extracted from the segmentations of the prostate gland in one patient's MRI and TRUS images respectively (Fig. 1). The second dataset included 75 MRI and TRUS point-set pairs for training and 33 pairs from different patients for testing, in order to validate the generalisability of the developed population-trained model. The third dataset contains $N_k = 22$ pairs of point sets generated over MRI-derived prostate meshes [35] by producing ground-truth deformations in $[5.58, 8.66]$ mm using the finite element modelling (FEM) process, proposed in previous studies [14], [31], with different material properties assigned to peripheral zones (PZ) and transition zone (TZ), and the ratios of Youngs' Modulus with PZ and TZ $\frac{E^{\text{PZ}}_k}{E^{\text{TZ}}_k}$ were in the range of $[0.10, 0.20]$. We follow the same ranges of material model as those in [31]. Specifically, the prostate tissues are assumed to be isotropic and elastic materials where the nonlinear relationship between applied stress and strain is characterised by a neo-Hookean model. In addition, the prediction of displacement in FEM depends only on the ratio of elasticity moduli assigned to different prostate zones [31]. The shear and bulk modulus for both sub-regions are sampled in the range of $[3.36, 76.9]$kPa and $[8.33, 3.33 \times 10^3]$kPa. In the third dataset, the data was resampled to isotropic resolutions being $1 \times 1 \times 1$ mm^3. In all three datasets, for each patient, the pairs of MRI and TRUS point sets were randomly downsampled to $\mathbf{P}_\mathcal{S}$ and $\mathbf{P}_\mathcal{T}$ with $N_s = N_t = 1024$ and $N^{\text{surface}}_s = N^{\text{surface}}_t = 512$ independently.

Overall, there are four experiments where the first three experiments of the forward problem that registers $\mathbf{P}_\mathcal{S}$ and $\mathbf{P}_\mathcal{T}$ were conducted on the first, second and third datasets while the fourth experiment of the inverse problem estimating the ratio of Young's Modulus between two prostate sub-regions (i.e., PZ and TZ) was conducted on the third dataset. In the first and second experiments, the adapted linear elasticity in Sect. II-B is used. In the third and fourth experiments where the ground-truth deformations are known in the utilised data set, we compare the adapted linear and nonlinear elasticities in Sect. II-B and Sect. III-A.

B. Evaluation Metrics

For the experimental results of the forward problem using the clinical data described in Sec. IV-D, four evaluation metrics are reported. First, target registration errors (TREs) were computed as the average distance between the geometric centroids of pairs of registered source and target landmarks, which include apex and base of the prostate, water-filled cysts, and calcifications [14], [31]. Further details in defining these independent landmarks followed published methods in previous studies [2], [4]. Second, deformation magnitudes (DMs) were computed to measure the "pure" non-rigid part of predicted displacements of $\mathbf{P}_\mathcal{S}$, excluding the "largest" rigid transformation $(\mathbf{R},\mathbf{t})$. DM was defined as residuals after solving the orthogonal Procrustes problem (i.e., the process of computing $\mathbf{R}$ and $\mathbf{t}$) between $\mathbf{P}_\mathcal{S}$ and $\mathsf{T}(\mathbf{P}_\mathcal{S})$ [16]: as

$$\text{DM} = \frac{1}{|\widetilde{N}|}\sum_{s\in\widetilde{N}} ||\mathbf{R}\mathbf{p}_s + \mathbf{t} - \mathsf{T}^\star(\mathbf{p}_s)||_2, \tag{24}$$

where $\widetilde{N}$ can be either $\mathbf{P}_\mathcal{S}$ or $\mathbf{P}^{\text{internal}}_\mathcal{S}$, $|\widetilde{N}|$ is the number of points, $\mathsf{T}^\star$ denotes the estimated non-rigid transformation with different registration aglorithms. Third, the Chamfer Distance (CD) was defined as

$$\begin{aligned}\text{CD} = \frac{1}{2}\Big(\frac{1}{|\widetilde{N}_t|}\sum_{t\in\widetilde{N}_t}\min_{s\in\widetilde{N}_s}||\mathsf{T}^\star(\mathbf{p}_s) - \mathbf{p}_t||_2 + \\ \frac{1}{|\widetilde{N}_s|}\sum_{s\in\widetilde{N}_s}\min_{t\in\widetilde{N}_t}||\mathsf{T}^\star(\mathbf{p}_s) - \mathbf{p}_t||_2\Big),\end{aligned} \tag{25}$$

where $\widetilde{N}_t$ and $\widetilde{N}_s$ are the same as those in Eq. (7). The principal stretches $\{\lambda^s_i \in \mathbb{R}^+\}^3_{i=1}$ for the s-th point $\mathbf{p}_s$ are computed as the positive eigenvalues of the left Cauchy-Green Tensor $\mathbf{F}_s\mathbf{F}^\mathsf{T}_s$ using the predicted displacement vector $\mathbf{d}_s$. We compute the relative stretch ratio between the two sub-regions of the prostate gland as

$$f(\lambda_i) = |\lambda^{\text{soft}}_i - 1|/|\lambda^{\text{rigid}}_i - 1|, \tag{26}$$

where $|\bullet|$ denotes the absolute value of a scalar, $\lambda^{\text{soft}}_i \in \mathbb{R}^+$ and $\lambda^{\text{rigid}}_i \in \mathbb{R}^+$ denote the i-th ($i \in \{1,2,3\}$) principal stretch in the soft and rigid sub-regions, respectively.

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For the experiments using simulated data, the root-mean-square error (rmse) was defined between predicted displacement $\mathbf{D}_{\mathcal{S}}$ and ground-truth $\mathbf{D}_{\mathcal{S}}^{gt} \in \mathbb{R}^{|\widetilde{N}_s|\times 3}$ as

$$\text{RMSE} = \sqrt{\frac{1}{|\widetilde{N}_s|}\sum_{s\in\widetilde{N}_s} ||\mathbf{d}_s - \mathbf{d}_s^{\text{gt}}||_2^2}, \tag{27}$$

where $\mathbf{d}_s^{\text{gt}} \in \mathbb{R}^3$ is the ground-truth displacement vector with $\mathbf{p}_s$. The mean absolute error (MAE) between the i-th principal stretches using predicted and ground-truth displacement vectors are computed as

$$\text{MAE}(\lambda_i) = \frac{1}{|\widetilde{N}_s|}\sum_{s\in\widetilde{N}_s} |\lambda_i^s - \lambda_i^{s,\text{gt}}|, \tag{28}$$

where $\lambda_i^{s,\text{gt}} \in \mathbb{R}^+$ is the i-th principal stretch computed using the ground-truth displacement vector $\mathbf{d}_s^{\text{gt}}$. To evaluate the performances with different models to solve the inverse problem in Section. IV-E, the absolute percentage error (APE) is defined as

$$\text{APE} = |(\text{ratio}_{\text{predict}} - \text{ratio}_{\text{gt}})/\text{ratio}_{\text{gt}}, \tag{29}$$

for each case where $\text{ratio}_{\text{predict}}$ and ratio_{gt} are predicted and ground-truth ratios $\frac{E_k^{\text{PZ}}}{E_k^{\text{TZ}}}$, and also reported mean APE (mAPE) for all involved cases.

C. Implementation Details

1) Network Details: PointNet [7] is adapted with a TNet 4-by-4 outputting a 4×4 rigid transformation matrix instead of the original 3-by-3 TNet, as suggested in [2]. The final global feature from a PointNet $\phi(\cdot)$ is of size 1024. In the global feature extraction (Fig. 1) module, the global features $\phi(\mathbf{P}_{\mathcal{S}})$ and $\phi(\mathbf{P}_{\mathcal{T}})$ learnt from $\mathbf{P}_{\mathcal{S}}$ and $\mathbf{P}_{\mathcal{T}}$ are concatenated. In the point transformation module (Fig. 1), the concatenated global feature is repeated for N_s times and further concatenated with $\mathbf{P}_{\mathcal{S}}$. The resulting feature map of size $N_s \times 2051$ will go through shared MLP(1024, 512, 256, 128, 64) and another shared MLP(256) without the ReLU layer. At the end, MLP(3) and 6 individual MLP(1) are used in branches $g_{\theta_g}(\mathcal{P}_k)$ predicting $\mathbf{D}_{\mathcal{S}}$ and $h_{\theta_h}(\mathcal{P}_k)$ predicting σ, respectively.

2) Parameter Settings: To formulate and compute the leveraged biomechanics-related terms in the loss function (cf. Eq. (8)-Eq. (13)), it requires the point-wise Lamé parameters $\lambda_s > 0$ and $\mu_s > 0$ whose values directly depend on the point-wise Young's Modulus $E_s > 0$ and Possion's ratio $v_s \in (0, 0.5)$. This ideally necessitates knowing exact material parameters of every point, which is impractical in real-world scenarios. According to [31], displacement predictions in the finite element analysis are dependent solely on the ratio of material properties across different sub-regions, rather than on the absolute values of those properties. Furthermore, considering the fact or prior biomechanical knowledge that the tissues in the PZ region are softer than those in the TZ region, we manually assign an empirical Young's modulus values to points in the PZ region that are same across all patients, while a larger modulus value is then assigned to the TZ by applying a fixed multiplier that is also consistent for all patients. For the first and second experiments, Young's modulus E in Eq. (11) and Eq. (18)/Eq. (19) was chosen as 500 kPa and 5 kPa for points in rigid and soft compartments while Possion' ratio v was 0.49, leading to $(\lambda = 8221.48, \mu = 167.78)$ and $(\lambda = 82.21, \mu = 1.68)$, respectively. For the third experiment, E and v were set according to the ratio of their ground-truth values in two sub-regions. All four experiments were run on an NVIDIA GeForce RTX 4090 GPU (24 GB memory) and an Intel Core i9-14900KF CPU.

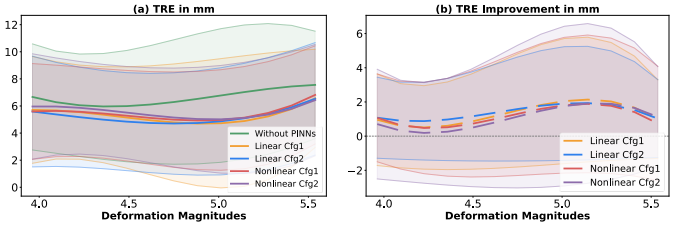


Fig. 2. (Left) TRE (mm) of MRI-TRUS fusion using patient-specific registration models with and without PINNs, w.r.t. deformation magnitudes of prostate gland. (Right) TRE improvements by incorporating PINNs compared to those without PINNs. The shaded areas are **95%** confidence intervals (i.e., $\pm$**2** standard deviations).

3) Compared Methods: We have compared our method with state-of-the-art registration approaches, including biomechanics-based methods such as GMM-FEM [9] and WarpPINNs [24], smoothness-based approaches including CPD [8] and Smooth-PINNs (cf. Appendix VII-E for details) where biomechanical constraints in our framework are replaced with smoothness-based regularizations [36], WithoutPINNs being a variant of our method that retains only the Chamfer loss, the data-driven method NDP [37] and the optimisation-based registration approach PointSetReg [38].

D. Results of the Forward Problem

1) Results of the Forward Problem using the Linear Elasticity under the Single-Pair Patient-Specific Training Paradigm: Table I shows the mean and standard deviation of the evaluation metrics including TRE, DM, CD and %neg J, in the first experiment. Three observations can be made from Table I: *1)* TRE values decreased with PINNs significantly for all four proposed PINNs-based variants ($p < 0.001$), compared against without PINNs; *2)* DM values in the rigid sub-regions were smaller than those in the soft sub-regions with PINNs, which demonstrated biomechanical constraints are effectively preserved in the registration algorithm, i.e., $\frac{\text{DM}_{\text{rigid}}}{\text{DM}_{\text{soft}}} < 1$, different from $\frac{\text{DM}_{\text{rigid}}}{\text{DM}_{\text{soft}}} > 1$ without PINNs, for Linear Cfg1 ($p = 0.15$), Linear Cfg2 ($p < 0.01$), Nonlinear Cfg1 ($p < 0.05$) and Nonlinear Cfg2 ($p = 0.30$) respectively; and *3)* %neg J is significantly reduced from 0.39 ± 0.56 without PINNs to 0.24 ± 0.53 ($p < 0.01$), 0.16 ± 0.25 ($p < 0.001$), 0.14 ± 0.28 ($p < 0.001$) and 0.15 ± 0.59 ($p < 0.01$) using PINNs-based registration variants. It is noted that the sample size in the paired t-test is 77. In addition, Linear Cfg2 significantly outperform all other compared methods in terms of both TRE ($p < 0.01$) and %neg J ($p < 0.001$).

Fig. 2 shows TRE values w.r.t. varying DM thresholds computed before registration. It is found from Fig. 2 that

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TABLE I

QUANTITATIVE RESULTS (MEAN ± STD IN MM) OF PATIENT-SPECIFIC MODELS IN THE FIRST EXPERIMENT. STATISTICAL SIGNIFICANCE IS INDICATED BY ASTERISKS: ***: $p < 0.001$, **:$0.001 < p < 0.01$, *:$0.01 < p < 0.05$, $^{\circ}$: STATISTICAL INSIGNIFICANCE AT $\alpha = 0.05$. IT IS NOTED THAT THE FIRST EIGHT ROWS AND THE LAST FOUR ROWS REPORT THE P-VALUES OF STATISTICAL TESTS COMPUTED BY COMPARING THE RESULTS WITH LINEAR CFG2 AND WITHOUT PINNS, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD FOR %NEG J, CD AND TRE.

Methods	DM (Internal Points Rigid Region)	DM (Internal Points Soft Region)	%neg J	CD	CD (Surface Points Only)	TRE
CPD [8]	$2.99 \pm 1.09^{***}$	$2.17 \pm 1.08^{***}$	$0.39 \pm 0.60^{***}$	$7.88 \pm 4.20^{***}$	$9.87 \pm 4.54^{***}$	$19.59 \pm 7.12^{***}$
PointSetReg [38]	$3.73 \pm 0.72^{***}$	$4.41 \pm 0.84^{***}$	$2.59 \pm 1.85^{***}$	$\mathbf{1.49 \pm 0.21}^{***}$	$1.78 \pm 0.32^{***}$	$6.06 \pm 2.45^{**}$
NDP [37]	$3.63 \pm 1.24^{***}$	$4.42 \pm 1.44^{***}$	$0.38 \pm 0.70^{***}$	$1.82 \pm 0.26^{***}$	$2.15 \pm 0.40^{***}$	$7.05 \pm 3.17^{***}$
GMM-FEM [9]	$8.55 \pm 2.52^{***}$	$10.02 \pm 2.88^{***}$	$15.96 \pm 5.33^{***}$	$3.49 \pm 0.71^{***}$	$5.09 \pm 1.23^{***}$	$10.18 \pm 6.15^{***}$
ICP+GMM-FEM [9]	$1.76 \pm 1.01^{***}$	$2.53 \pm 0.97^{***}$	$0.95 \pm 1.90^{***}$	$2.44 \pm 0.43^{***}$	$2.80 \pm 0.62^{***}$	$6.48 \pm 2.76^{***}$
Smooth-PINNs [36]	$4.72 \pm 1.04^{***}$	$5.29 \pm 1.16^{***}$	$0.42 \pm 0.83^{***}$	$1.76 \pm 0.26^{\circ}$	$1.13 \pm 0.17^{\circ}$	$7.04 \pm 2.38^{***}$
WarpPINN [24]	$4.92 \pm 1.12^{***}$	$5.50 \pm 1.24^{***}$	$0.58 \pm 0.96^{***}$	$1.50 \pm 0.26^{***}$	$\mathbf{0.53 \pm 0.41}^{***}$	$7.04 \pm 2.27^{***}$
Without PINNS [2]	$4.78 \pm 1.06^{***}$	$4.72 \pm 0.95^{***}$	$0.39 \pm 0.56^{***}$	$1.65 \pm 0.25^{\circ}$	$0.86 \pm 0.15^{\circ}$	$6.92 \pm 2.32^{***}$
Linear Cfg1	$4.38 \pm 0.97^{***}$	$4.92 \pm 1.09^{***}$	$0.24 \pm 0.53^{**}$	$1.64 \pm 0.24^{\circ}$	$0.84 \pm 0.17^{\circ}$	$5.67 \pm 2.14^{***}$
Linear Cfg2	$4.28 \pm 0.90^{***}$	$4.83 \pm 1.04^{***}$	$0.16 \pm 0.25^{***}$	$1.67 \pm 0.31^{\circ}$	$0.90 \pm 0.38^{\circ}$	$\mathbf{5.58 \pm 2.24}^{***}$
Nonlinear Cfg1	$4.89 \pm 1.06^{**}$	$5.46 \pm 1.17^{**}$	$\mathbf{0.14 \pm 0.28}^{***}$	$1.54 \pm 0.24^{***}$	$0.61 \pm 0.22^{***}$	$5.83 \pm 2.00^{***}$
Nonlinear Cfg2	$4.78 \pm 0.95^{\circ}$	$5.36 \pm 1.09^{\circ}$	$0.15 \pm 0.59^{**}$	$1.55 \pm 0.27^{***}$	$0.62 \pm 0.31^{***}$	$5.96 \pm 2.25^{***}$

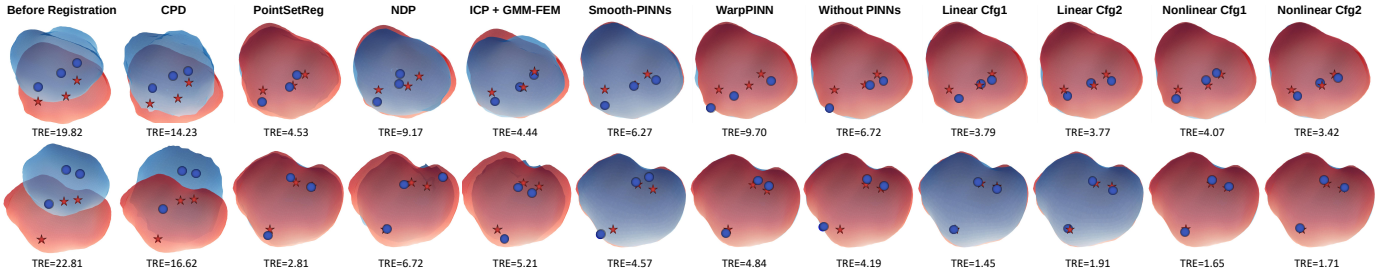


Fig. 3. Qualitative comparisons of all methods in the first experiment. The original and warped source (i.e., MRI) meshes are depicted in blue while target (i.e., TRUS) meshes are in red. The anatomical landmarks in MRI and TRUS are denoted by blue circles and red stars. TRE values are shown underneath each sub-figure.

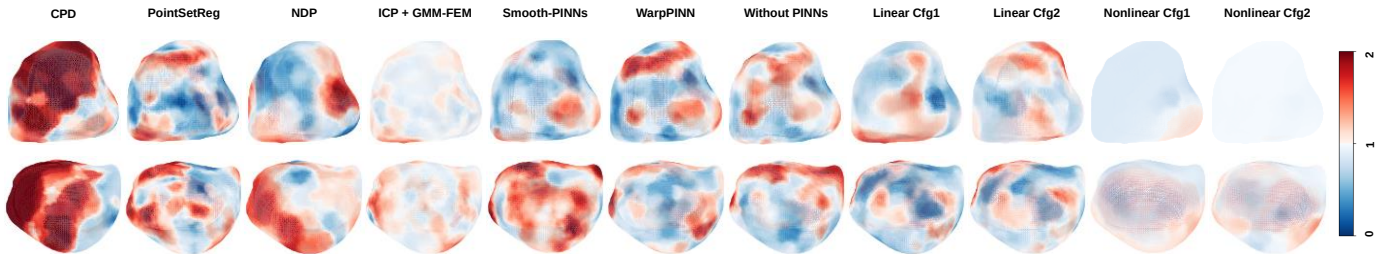


Fig. 4. The qualitative comparison of Jacobian determinants computed using deformation fields estimated with different registration algorithms in the first experiment. The Jacobian determinant (i.e., $\boldsymbol{J} = \det \mathbf{J}$) of the transformation is mapped onto the ultrasound surface using a color map. Deep-blue represents maximum local compression (i.e., $\boldsymbol{J} \approx \mathbf{0}$), white indicates volume preservation (i.e., $\boldsymbol{J} = \mathbf{1}$), and deep-red denotes maximum local expansion (i.e., $\boldsymbol{J} \approx \mathbf{2}$). Ideally, the deformation should yield $\boldsymbol{J}$ around unity ($\boldsymbol{J} = \mathbf{1}$) and maintain a spatially smooth transformation field.

TABLE II

TRE STATISTICS (MEAN ± STD IN MM) OF PATIENTS WHOSE RATIOS OF PRINCIPAL STRETCH BETWEEN SOFT AND RIGID SUB-REGIONS (CF. EQ. (26)) WERE CORRECTLY MODULATED FROM $< \mathbf{1}$ WITHOUT PINNS TO $> \mathbf{1}$ WITH PINNS. (1) THE $\mathbf{1, 3, 5..., 11}^{th}$ ROWS DENOTE SUCH CASES; AND (2) THE $\mathbf{2, 4, 6..., 12}^{th}$ ROWS DENOTE PATIENTS IN (1) WHOSE TRE WERE ALSO IMPROVED. THE COLUMN # RECORDS THE NUMBER OF PATIENTS PER ROW. ***: $p < 0.001$, **:$0.001 < p < 0.01$, $^{\circ}$: STATISTICAL INSIGNIFICANCE AT $\alpha = 0.05$.

Methods	#	TRE Without PINNs	TRE With PINNs	TRE Improved With PINNs
Smooth-PINNs	18	6.17 ± 2.82	$\mathbf{6.15 \pm 2.08}^{\circ}$	0.03 ± 1.93
Smooth-PINNs	8	8.62 ± 2.37	$\mathbf{6.95 \pm 2.08}^{**}$	1.67 ± 1.21
WarpPINN [24]	22	7.14 ± 2.38	$\mathbf{6.85 \pm 2.19}^{\circ}$	0.28 ± 1.80
WarpPINN [24]	11	8.42 ± 1.80	$\mathbf{6.80 \pm 2.18}^{**}$	1.62 ± 1.58
Linear Cfg1	25	6.36 ± 1.92	$\mathbf{5.18 \pm 2.01}^{***}$	1.18 ± 1.40
Linear Cfg1	17	6.70 ± 1.91	$\mathbf{4.68 \pm 2.01}^{***}$	2.02 ± 0.79
Linear Cfg2	27	6.32 ± 2.90	$\mathbf{4.84 \pm 2.58}^{***}$	1.48 ± 1.42
Linear Cfg2	22	6.65 ± 2.95	$\mathbf{4.71 \pm 2.63}^{***}$	1.94 ± 1.15
Nonlinear Cfg1	34	6.83 ± 2.60	$\mathbf{5.59 \pm 2.22}^{***}$	1.24 ± 1.75
Nonlinear Cfg1	24	7.51 ± 2.62	$\mathbf{5.39 \pm 2.14}^{***}$	2.12 ± 1.18
Nonlinear Cfg2	29	6.39 ± 1.88	$\mathbf{5.66 \pm 1.73}^{\circ}$	0.73 ± 1.90
Nonlinear Cfg2	20	7.13 ± 1.61	$\mathbf{5.52 \pm 1.76}^{***}$	1.61 ± 1.46

TABLE III

THE RATIO OF PRINCIPAL STRETCHES BETWEEN THE SOFT AND RIGID SUB-REGIONS ALONG THREE MUTUALLY PERPENDICULAR DIRECTIONS (I.E., THE COLUMNS $\boldsymbol{f(\lambda_1)}$,$\boldsymbol{f(\lambda_2)}$ AND $\boldsymbol{f(\lambda_3)}$), AND THE LOCAL VOLUME CHANGE RATE (I.E., THE COLUMN $\boldsymbol{\lambda_1 \cdot \lambda_2 \cdot \lambda_3}$).

Methods	$f(\lambda_1)$	$f(\lambda_2)$	$f(\lambda_3)$	$\lambda_1 \cdot \lambda_2 \cdot \lambda_3$
CPD [8]	1.01 ± 1.18	0.45 ± 0.56	0.60 ± 0.21	1.52 ± 0.93
PointSetReg [38]	0.94 ± 0.18	2.28 ± 2.74	1.19 ± 0.29	1.55 ± 1.31
NDP [37]	1.08 ± 0.31	1.92 ± 2.58	1.16 ± 0.37	1.27 ± 0.82
GMM-FEM [9]	0.68 ± 0.10	1.95 ± 2.28	1.20 ± 0.32	4.14 ± 4.95
ICP+GMM-FEM [9]	1.23 ± 0.46	1.66 ± 2.19	0.96 ± 0.42	1.09 ± 0.41
Smooth-PINNs [36]	1.26 ± 1.14	1.14 ± 1.65	1.07 ± 0.27	1.50 ± 1.26
WarpPINN [24]	1.17 ± 0.83	1.43 ± 2.02	1.08 ± 0.22	1.51 ± 1.33
Without PINNs [2]	1.12 ± 0.39	0.71 ± 0.43	1.06 ± 0.22	1.44 ± 1.18
Linear Cfg1	1.45 ± 1.63	1.42 ± 1.82	1.09 ± 0.27	1.21 ± 0.79
Linear Cfg2	1.40 ± 1.49	1.55 ± 2.22	1.14 ± 0.28	1.31 ± 0.92
Nonlinear Cfg1	1.54 ± 1.63	1.45 ± 1.65	1.17 ± 0.51	1.05 ± 0.38
Nonlinear Cfg2	1.14 ± 0.61	1.52 ± 2.03	1.25 ± 1.09	$\mathbf{1.04 \pm 0.45}$

1) TRE values increased with larger deformation magnitudes for all methods; and *2)* PINNs reduced the TRE values and demonstrated greater improvements for patients that undergo larger deformations. For example, Linear Cfg1, Linear Cfg2,

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Nonlinear Cfg1 and Nonlinear Cfg2 significantly ($p < 0.001$) decreased TRE from 7.86 ± 2.43 mm without PINNs to 6.25 ± 2.38, 6.35 ± 2.50, 6.42 ± 2.18 and 6.40 ± 2.71, among top 40% (31/77) patients with larger deformations.

Fig. 3 shows qualitative results of two specific patients (*Case 1* and *Case 2*), with large and moderate-to-large non-rigid deformations being 6.01 mm and 4.65 mm. Taking *Case 1* as an example, the PINNs-based registration methods achieved the desired results: smaller DM values were found in the rigid sub-region than in the soft sub-region. More specifically, the DM values in the rigid and soft sub-regions are 4.84 mm versus 5.82 mm for Linear Cfg1, 4.70 mm vs 5.65 mm for Linear Cfg2, 6.04 mm versus 6.74 mm for Nonlinear Cfg1, and 5.55 mm vs 6.47 mm for Nonlinear Cfg2, whereas without PINNs DM was larger (i.e., 5.44 mm) in the rigid compartment than that (i.e., 4.58 mm) in the soft one. Moreover, the second relative stretch ratios $f(\lambda_2)$ in Eq. (26) between the two sub-regions of the prostate gland are 1.28, 1.13, 0.60 and 0.97 for the four PINNs-based variants respectively, desirably better than that being 0.38 without PINNs. While surface points are visually well aligned using all methods (cf. Fig. 3) with Chamfer distances being 0.68 mm, 0.81 mm, 0.59 mm, 0.63 mm and 0.88 mm for Linear Cfg1, Linear Cfg2, Nonlinear Cfg1, Nonlinear Cfg2, and without PINNs respectively, PINNs greatly reduced TRE from that without PINNs, i.e., from 6.72 mm to 3.79 mm (Linear Cfg1), 3.77 mm (Linear Cfg2), 4.07 mm (Nonlinear Cfg1) and 3.42 mm (Nonlinear Cfg2), which demonstrates the effectiveness of PINNs in producing more clinically meaningful deformations. In addition, as shown in Fig. 3, all four proposed PINNs-based variants achieve lower TRE values and better alignment of prostate surfaces than other compared methods.

Table II shows the mean and standard deviation of TRE for those specific patients whose ratios of principal stretch between soft and rigid sub-regions were correctly modulated from < 1 without PINNs to > 1 with PINNs. As shown in Table II, 25 (*Linear Cfg1*), 27 (*Linear Cfg2*), 34 (*Nonlinear Cfg1*) and 29 (*Nonlinear Cfg2*) achieved desired smaller principal stretch in the rigid sub-regions than those in the soft sub-regions, while the principal stretches for those cases were larger in the rigid sub-regions than those in the soft sub-regions without PINNs. The majority, being 68% (17/25), 82% (22/27), 71% (24/34) and 69% (20/29) obtained lower TREs than those without PINNs, for the four proposed PINNs-based variants, where TRE improvements were statistically significant ($p < 0.001$) with mean differences being 2.02 mm, 1.94 mm, 2.12 mm and 1.61 mm, respectively. This is consistent with conclusions from previous studies, which have shown that imposing distinct material properties within the registration improves accuracy. Similar facts can be observed across other compared PINNs-based registration methods including Smooth-PINNs and WarpPINN [24].

Table III shows the relative stretch ratio between the two sub-regions of the prostate gland, computed using Eq. (26). Two observations can be found by comparing results without PINNs with PINNs-based registration variants in Table III: *1)* the stretch ratios were improved with PINNs particularly along the 2nd principal stretch, i.e., $f(\lambda_2)$ were changed from the 'smaller-than-one' value being 0.71 ± 0.43 without PINNs to the desired 'greater-than-one' values being 1.42 ± 1.82 (*Linear Cfg1*), 1.55 ± 2.22 (*Linear Cfg2*), 1.45 ± 1.65 (*Nonlinear Cfg1*) and 1.52 ± 2.03 (*Nonlinear Cfg2*); *2)* the product of principal stretches $\lambda_1 \cdot \lambda_2 \cdot \lambda_3$ being closer to 1 using PINNs than that without PINNs exhibits PINN's superior volume-preserving capability, e.g., being 1.04 ± 0.45 vs 1.44 ± 1.18 for Nonlinear Cfg2 and without PINNs, respectively. As it is shown in Table III, while NDP [37], Smooth-PINNs and WarpPINNs [24] also achieve larger-than-one values of the relative ratios for all three principal stretches, CPD [8] achieves a smaller-than-one (not desired) values for $f(\lambda_2)$ and $f(\lambda_3)$. In addition, the product of principal stretches $\lambda_1 \cdot \lambda_2 \cdot \lambda_3$ is closer to 1 using proposed PINNs-based registration variants than most compared methods, exhibiting their superior volume-preserving capabilities.

TABLE IV

MEAN ABSOLUTE ERROR BETWEEN THE GROUND TRUTH AND PREDICTION FOR DEFORMATION FIELD, AND PRINCIPAL STRETCHES IN THE THREE PERPENDICULAR DIRECTIONS. THE LAST ROW SHOWS THE PEARSON CORRELATION COEFFICIENTS (PCCS) BETWEEN PRINCIPAL STRETCHES ($\boldsymbol{\lambda_1, \lambda_2, \lambda_3}$) DIFFERENCES AND RMSE.

Methods	RMSE	MAE(λ_1)	MAE(λ_2)	MAE(λ_3)
Smooth-PINNs [36]	1.88°	$0.113^{\star}$	0.022°	$0.122^{\star}$
WarpPINN [24]	1.87°	$0.115^{\star}$	0.022°	$0.126^{\star\star}$
Without PINNs [2]	1.80	0.101	0.021	0.111
Linear Cfg1	1.51°	$0.061^{\star\star}$	$\mathbf{0.013}^{\circ}$	$0.065^{\star\star\star}$
Linear Cfg2	$1.27^{\star\star}$	$\mathbf{0.049}^{\star\star\star}$	0.014°	$0.056^{\star\star\star}$
Nonlinear Cfg1	$1.51^{\star}$	$0.056^{\star\star\star}$	0.016°	$0.056^{\star\star\star}$
Nonlinear Cfg2	$1.28^{\star\star\star}$	$0.050^{\star\star\star}$	$\mathbf{0.013}^{\star}$	$\mathbf{0.051}^{\star\star\star}$
Pearson Correlation Coefficient	-	0.850	0.730	0.915

Fig. 4 shows the qualitative comparison of Jacobian determinants computed using deformation fields computed using different registration methods. As shown in Fig. 4, the proposed PINNs-based variants, particularly Nonlinear Cfg1 and Nonlinear Cfg2, exhibit notably lighter and smoother color spatial distributions compared to most compared approaches (especially Without PINNs). These visual differences suggest that the PINNs-based methods better satisfy biomechanical constraints and achieve superior volume preservation. These findings align with the common biomechanical assumption that the biological tissues are nearly incompressible, though we acknowledge that this property may not be exactly the case for prostate [1]).

2) Results of the Forward Problem using the Linear Elasticity under the Multi-Patient Training Paradigm: For the second experiment, compared to that without PINNs, the incorporation of PINNs ($w = 10^3$) significantly reduced the average TREs on the test subjects from 6.96 ± 1.90 mm to 6.12 ± 1.95 mm ($p = 0.018$), while Chamfer distances with and without PINNs were 2.48 ± 0.33 mm and 2.54 ± 0.37 mm ($p = 0.165$) on all points (2.96 ± 0.55 mm and 2.60 ± 0.41 mm ($p < 0.001$) on surface points), respectively. The slightly larger CD values of surface points using PINNs than those without PINNs imply that additional biomechanical constraints are successfully learnt and encoded by the network and take effect in deforming the MRI point set. At the same time, the significant improved TRE values verify benefits of PINNs in producing more plausible deformations on unseen test patients.

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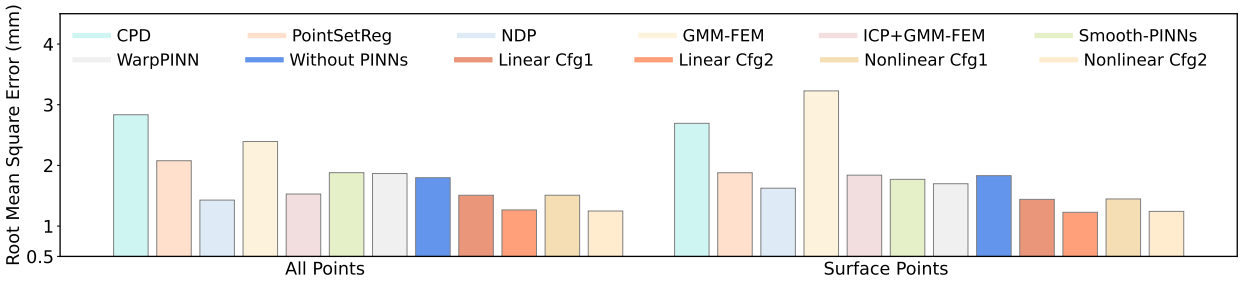


Fig. 5. The root-mean-squared-error (rmse) of registration in the third experiment, using different algorithms for surface and all points on the left and right subplots, respectively.

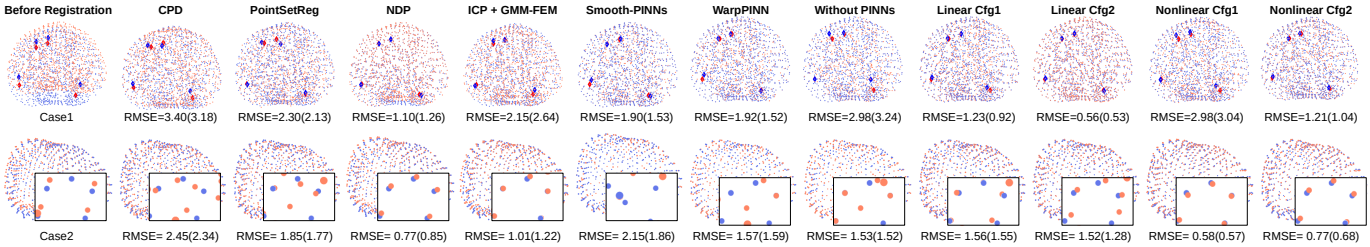


Fig. 6. Qualitative results for two cases with the compared registration methods. The rmse values for all points are also reported followed by those in parentheses for surface points.

The successful imposition of biomechanical constraints on the test data was further demonstrated by *1)* The ratios of DM between internal points in rigid and soft compartments $\frac{\text{DM}_{\text{rigid}}}{\text{DM}_{\text{soft}}}$ were 0.89 ± 0.11 and 1.35 ± 0.15 $(p < 0.001)$ using registration methods with and without PINNs, respectively; and *2)* The loss computed on the test patients using Eq. (14) was reduced from 20 to 10^{-14} after registration, which demonstrated the network's strong ability of inferring constraints on unseen subjects. The value of $\frac{\text{DM}_{\text{rigid}}}{\text{DM}_{\text{soft}}}$ being smaller than one using PINNs, against that being larger than one without PINNs, demonstrates the (more) incompressibility characteristic of tissues in the rigid region is clearly guaranteed. Furthermore, the physics-constrained loss terms, minimized on the training data, reach very small values on unseen test patients. To summarise, the underlying elasticity physics are successfully learned by the network and generalized to produce the expected positive effect when registering unseen test cases.

3) Impact of Nonlinear Models on Solving the Forward Problem: Fig. 5 includes the RMSE values computed using Eq. (27) for all (left) and surface (right) points respectively in the third experiment. Several key observations can be made from Fig. 5: (1) except for the case with Linear Cfg1 for all points ($p = 0.083$), all PINNs-assisted methods outperformed Without PINNs significantly at $\alpha = 0.05$ level; (2) except for the case where Linear Cfg1 and Linear Cfg2 were significantly different for all points ($p = 0.0496$), the differences between every pair of PINNs algorithms were not statistically significant; (3) Nonlinear Cfg2 achieved the lowest mean registration error values being 1.25 ± 0.62 mm and 1.24 ± 0.68 mm for both all and surface points ($p < 0.001$), against 1.80 ± 0.44 mm and 1.83 ± 0.51 mm Without PINNs. Fig. 5 further shows that Linear Cfg2 and Nonlinear Cfg2, achieve lower registration error values than all compared methods in terms of all points. In addition, all four proposed PINNs-based registration approaches consistently outperform all benchmark methods specifically for surface points. It is expected that the differences between linear and nonlinear PINNs-based registration variants are not significant, given that the deformation magnitude before registration being sampled in $[5.58, 8.66]$ mm is relatively small (cf. Fig. 6). This conclusion should be taken carefully, particularly in cases where both models are applied to different datasets with possibly distinct deformation magnitudes from tested values in this paper.

Fig. 6 shows the registration performances with two specific patient cases (i.e., *Case 1* and *Case 2*) in the third experiment, where here blue and red dots respectively denote source (or predicted warped source) and target point sets with exact correspondences before (or after) registration. In *Case 1* (cf. the first row in Fig. 6), the linear model (e.g., rmse values were 0.56 mm and 0.53 mm for all and surface points using Linear Cfg2) was better than the nonlinear model (e.g., rmse values were 1.21 mm and 1.04 mm using Nonlinear Cfg2) and Without PINNs (i.e., rmse values were 2.98 mm and 3.24 mm), while in *Case 2* (cf. the second row in Fig. 6) nonlinear models outperformed both linear ones and Without PINNs. In the first row of Fig. 6, corresponding targets in two spaces to be registered are denoted with blue and red star shape. Fig. 6 further shows that 1) Linear Cfg2 yields lower registration errors values than all compared methods in terms of all and surface points for *Case 1*; 2) Nonlinear Cfg1 and Nonlinear Cfg2 consistently outperform all benchmark methods in terms of both all and surface points for *Case 2*.

To quantitatively evaluate the deformation field from the perspective of physical fidelity and biomechanical constraints, the mean absolute error between the ground-truth and the predicted main principal stretches are also computed (cf. $\text{MAE}(\lambda_i)$ in Eq. (28)). Table IV shows the corresponding results. The observations are summarised as follows. *1)* Among all PINNs-based methods, Linear Cfg2, Nonlinear Cfg1 and

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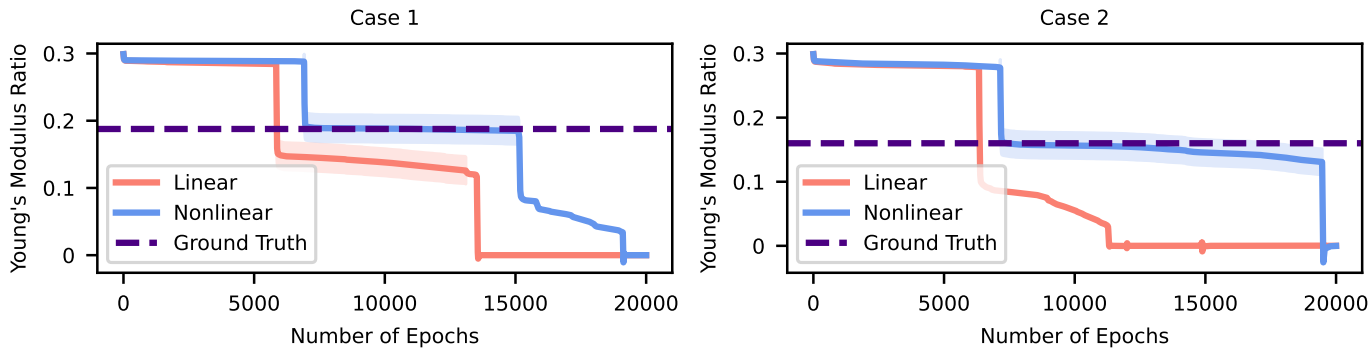


Fig. 7. The optimisation process of solving the inverse problem defined in Eq. (23) estimating the young's modulus ratio between two sub-regions of prostate for two specific patient cases (i.e., *Case 1* and *Case 2*), are found by locating saddle points. The estimated values using nonlinear and linear models are denoted with blue and red lines.

TABLE V
YOUNG'S MODULUS RATIO BETWEEN THE TWO REGIONS ESTIMATED WITH TWO MODELS FOR SIX PATIENT CASES.

Patient ID	Ground-truth Ratio	Predicted Ratio Linear	APE Linear	Predicted Ratio Nonlinear	APE Nonlinear
Case 1	0.19	0.12	36.84%	**0.19**	**1.23%**
Case 2	0.16	0.10	37.50%	**0.16**	**1.25%**
Case 3	0.11	0.22	100.00%	**0.10**	**9.09%**
Case 4	0.12	0.22	83.33%	**0.15**	**25.00%**
Case 5	0.17	0	100.00%	**0.19**	**11.76%**
Case 6	0.19	0	100.00%	**0.12**	**36.84%**

Nonlinear Cfg2 achieved significant reductions in RMSE compared against Without PINNs ($p < 0.05$). In contrast, Smooth-PINNs ($p = 0.56$) and WarpPINN ($p = 0.59$) did not show significant differences in RMSE over Without PINNs. *2)* For the first main principal stretch λ_1, compared against without PINNs, the proposed PINNs-based methods showed significant reductions ($p < 0.01$), whereas Smooth-PINNs ($p < 0.05$) and WarpPINN ($p < 0.05$) achieved slightly larger error values. *3)* For λ_2, Linear Cfg1 and Nonlinear Cfg2 achieve significant better results ($p < 0.05$) while Smooth-PINNs ($p = 0.38$), WarpPINN ($p = 0.51$), Linear Cfg2 ($p = 0.07$) and Nonlinear Cfg1 ($p = 0.06$) achieved slightly larger error values but without statistical significance. *4)* The third principal stretch λ_3 was consistently and significantly reduced by the proposed PINNs-based methods ($p < 0.001$) while Smooth-PINNs ($p = 0.027$) and WarpPINN ($p < 0.01$) exhibit significant larger error values. Lastly, in Table IV we report the Pearson Correlation Coefficient (PCC) between RMSE and MAE(λ_i). As it is shown in Table IV, strong correlations exist between the two types of error values.

E. Results of the Inverse Problem

Fig. 7 plots the estimated ratios of Young's Modulus between PZ and TZ (i.e., $\frac{E_k^{\text{PZ}}}{E_k^{\text{TZ}}}$) for two specific patients (i.e., *Case 1* and *Case 2*) with the numbers of training epochs, by solving the inverse problem defined in Eq. (23). Fig. 7 demonstrates that the ratios of young's modulus between the PZ and TZ were recovered (only Cfg1s are reported due to much better performances than Cfg2s), by locating saddle points (i.e., the flat lines) during the optimisation process. As shown in Fig. 7, the nonlinear model (denoted with blue color) exhibited significant advantages over the linear counterparts (denoted with red color) which is expected because more subtle (high-order) information is preserved.

Table V shows the ground-truth and predicted ratios for successful cases, from which the APE was computed. The further calculated mAPE values with linear and nonlinear models were $76.28\% \pm 30.97\%$ and $14.20\% \pm 14.12\%$ respectively, evidently indicating the nonlinear model performed much better in solving the inverse problem ($p = 0.002$).

As we have discussed in Sec. III-C, though the inverse problem of estimating the material property (i.e., $\frac{E_k^{\text{PZ}}}{E_k^{\text{TZ}}}$) is ill-posed, the proposed method elaborated in Sec. III-C exhibits great potentials for identifying patient-specific physical properties. On one hand, by locating the saddle points during the optimisation process, we have demonstrated that this specific material property is identifiable in a subset of patients. On the other hand, as shown in Table V, the estimation of material property using the nonlinear elasticity model is accurate for these identifiable cases, of which the biomechanical property that the outer gland is more compliant than the inner gland is successfully discovered [31]. Nevertheless, we acknowledge that the question we raise in Sec. I-A.3 necessitates more future investigations, given that only one practical solution has been examined.

F. Ablation Studies

We have done additional ablation experiments where one proposed registration approach is compared to the corresponding variant in which either the elastic energy or the static equilibrium equation is utilised. Table VI shows results the RMSE values using Linear Cfg2 and Nonlinear Cfg2, against those using corresponding variants with either the elastic energy or the residuals of static equilibrium equation, in terms of all and surface points. As clearly shown in Table VI, for all test cases, combining both terms achieves smaller registration error values than those counterparts using either one individual term. For example, the mean registration error value for all points using Linear Cfg2 is 1.27 ± 0.67 mm, which is smaller than those being 1.52 ± 0.55 and 1.42 ± 0.56 with Linear

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TABLE VI
THE QUANTITATIVE RESULTS (I.E., RMSE IN EQ. (27)) OF ABLATION STUDIES COMPARING LINEAR CFG2 AND NONLINEAR CFG2 AND THEIR COUNTERPARTS WHERE EITHER ELASTIC ENERGY OR RESIDUALS OF STATIC EQUILIBRIUM EQUATION IS UTILISED.

	Elastic Energy	Static Equilibrium	Linear Cfg2
All Points	1.52 ± 0.55	1.42 ± 0.56	$\mathbf{1.27 \pm 0.67}$
Surface Points	1.40 ± 0.57	1.40 ± 0.59	$\mathbf{1.23 \pm 0.69}$
	Elastic Energy	Static Equilibrium	Nonlinear Cfg2
All Points	1.43 ± 0.50	1.39 ± 0.58	$\mathbf{1.28 \pm 0.62}$
Surface Points	1.30 ± 0.57	1.34 ± 0.66	$\mathbf{1.24 \pm 0.68}$

TABLE VII
COMPARISON OF GPU MEMORY CONSUMPTION, THE TOTAL NUMBER OF THE MODEL'S PARAMETERS, GFLOPS, AND TRAINING TIME ACROSS DIFFERENT LEARNING-BASED REGISTRATION METHODS.

Methods	Memory (MB)	# of Params (M)	GFLOPS	Training Time (h)
NDP [37]	14.06	0.31	0.63	1.7
Smooth-PINNs [36]	679.74	5.6	7.5	2.03
WarpPINN [24]	678.57	5.6	7.5	1.23
Without PINNs [2]	226.00	5.6	7.5	1.19
Linear Cfg1	1557.29	25.2	43.64	1.22
Linear Cfg2	1149.69	8.4	13.23	1.17
Nonlinear Cfg1	1378.60	25.2	43.64	1.48
Nonlinear Cfg2	1147.52	8.4	13.23	1.52

CFg2 (Elastic Energy) and Linear Cfg2 (Static Equilibium), respectively.

G. Computational Costs

The peak GPU memory consumption, the total number of the model's parameters, Giga Floating-Point Operations Per Second (GFLOPS), and training time of PINNs-based registration variants (i.e., Smooth-PINNs, WarpPINN, Linear Cfg1, Linear Cfg2, Nonlinear Cfg1 and Nonlinear Cfg2) and learning-based registration methods (i.e., NDP and Without PINNs) are summarised in Table VII. Several observations can be drawn from Table VII. First, the PINNs-based methods exhibit higher peak GPU memory consumption- approximately 5 to 6.9 times greater than that of Without PINNs method. This increase is primarily attributed to the additional network branch that estimates the point-wise stress tensors. In addition, the first network configurations (i.e., Linear Cfg1 and Nonlinear Cfg1) require more peak memory assumptions, have more parameters and demand higher GFLOPS compared to the second network configurations (i.e., Linear Cfg2 and Nonlinear Cfg2) respectively. This differene arises because Cfg1 employs two individual branches and Cfg2 uses only one shared branch to extract features. Lastly, we well note that the differences of training time between Without PINNs and PINNs-based registration variants are marginal, being 1.19h, 1.22h, 1.17h, 1.48h and 1.52h respectively. It is also worth emphasizing that the absolute values of peak GPU memory consumption and computing performances (i.e., 1.56 GB and 43.64 GFLOPS respectively) among four registration variants remain relatively low, considering the available memory and computing capacities (i.e., 24GB and 82.58 TFLOPS respectively) of the modern GPU used in this study. All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU (24 GB memory) and an Intel Core i9-14900KF CPU.

V. DISCUSSIONS

This study adapts for the multi-modality point-based registration, the governing equations in the form of PDEs involved in linear and nonlinear elasticities originating from the elastic mechanics theory. These adapted PDEs are further utilised and reformulated to define the biomechanics-imposed forward problem of predicting the point-wise displacement vectors (i.e., data-driven solutions of PDEs) and the inverse problem of jointly estimating the displacements and physics-related parameters (i.e., data-driven discovery of PDEs), where the unknowns are learnt with neural networks (e.g., PointNet and MLP). The results on the clinical MRI-TRUS prostate data (i.e., the first and second data-sets) presented in Sect. IV-D.1 and Sect. IV-D.2 demonstrates the effectiveness of the proposed PINNs-based registration method to simultaneously enhance the registration accuracy (e.g., lower TRE values, cf. Fig. 2 and Fig. 3, Table I and II) and impose the biomechanical constraints that the desired unknown deformations should satisfy (e.g., desired DM ratios, cf. Table I). In addition, the generalisabiity of the developed model to unseen subjects is also verfied (cf. Sect. IV-D.2). Lastly, we compare the capabilities of the linear and nonlinear elasticities in solving the forward and inverse problems in Sect. IV-D.3 and Sect. IV-E, using the third data-set with known FEM-generated ground-truth deformations. The results validate that there are no statistically significant differences between the two models in solving the forward problem (cf. Fig. 5 and 6), but the nonlinear model exhibits an evident advantage in the inverse problem (cf. Table V and Fig. 7).

It is noteworthy that smoothness-based and biomechanical-based criteria are two distinct regularization approaches: the former aims to ensure the spatial smoothness while the latter seeks to enforce the desired biomechanical constraints for predicted deformations. The rationale of choosing biomechanical constraints over smoothness-based regularization counterparts is as follows. First, the evident differences of material properties associated with two individual sub-regions of the prostate gland can be explicitly modeled with the biomechanical constraints represented by the set of elastic equations. Second, the over-simplified spatial smoothnesss regularization is insufficient to provide enough constraints and model the complex deformation of the prostate [8], [15], [20]. Third, the nature of prostate deformations (probably due to pressure from surrounding organs, the pubic bone and TRUS probe) encountered in real-world clinical scenarios is essentially biomechanical during many prostate interventions [1], [9], [31], [39], [40].

A. Limitations

This paper needs to be read with several limitations. First, the use of PINNs does not circumvent all limitations of biomechanical modelling shared with other approaches, such as assumptions of potentially subject-specific material properties. However, as we have demonstrated in Sect. IV-E, this opens up new opportunities for finding solutions to the material property estimation challenge, by formulating an inverse data-driven discovery problem of PDEs potentially approachable

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with PINNs [34]. Second, our validation is focused on the MRI-TRUS prostate registration, while it is of broad interest to explore the approach's effectiveness for wider clinical applications such as accurate and reliable myocardial motion tracking from cardiac cine MRI sequences [19]. Third, the drawn conclusions may not be directly generalizable given the limited data size (cf. Sect. IV-A). Fourth, although results (cf. Sec. IV-E) shed light on tackling the inverse problem of estimating material property along with the registration, it should be acknowledged that the proposed solution to solve the inverse problem (cf. Sec. III-C) is primarily practical in nature. In this regard, it requires further extensive efforts to (1) explore the inherent rationale and enhance the explainability of the solution, (2) improve success rates of estimating the physics-related parameters, for example, through adding additional regularization terms (e.g., priors) of the parameters. Fifth, although we have successfully verified the effectiveness of the proposed PINNs-based registration methods using common material property values across all patients, we believe the patient-specific parameters should be used and may achieve better results if they can be accurately estimated or measured in vivo in the future. Sixth, it is acknowledged that the estimated rigid transformation matrix between source point set $\mathbf{P}_{\mathcal{S}}$ and deformed source point set $\mathsf{T}(\mathbf{P}_{\mathcal{S}})$is not free from error. As a result, the subsequently computed deformation map (DM) in Eq. (24) may not perfectly reflect the true deformation of $\mathbf{P}_{\mathcal{S}}$. Therefore, the observations or conclusions drawn from the DM should be interpreted with caution. Nevertheless, it is noted that the rigid transformation estimated by solving the orthogonal Procrustes problem is the best we can acquire under the real scenarios without knowing the spatial information of corresponding rigid structures in two spaces. Last but not least, the influences of the nonlinear elasticity compared to the linear counterpart on the biomechanics-informed non-rigid point set registration performances, should be validated on larger diverse real-world and simulated datasets, encompassing a wider variety of organs and a broader range of deformations.

VI. Conclusion

To conclude, in this paper, we have utilised PINNs to non-rigidly register multi-modalities and estimate material properties of soft tissues, which are formulated as the forward (i.e., data-driven solutions of PDEs) and inverse (i.e., data-driven discovery of PDEs) problems respectively, considering both linear and more complex nonlinear elasticities. Experimental results on FEM-produced data and clinical MRI-TRUS paired image data, using both patient-specific and multi-patient learning models, demonstrated that the proposed framework is capable of lowering registration errors with presubscribed biomechanical characteristics and generalizability, promising for clinical use and wider research in PINN-based modelling. The validity of adopting PINNs for solving estimating parameters, together with the superiority of the nonlinear model, is also demonstrated. These conclusions are drawn based on clinical data from prostate cancer patients, for topical applications including intraoperative motion modelling and multimodal image registration, potentially, new applications in better characterisation of pathological tissue motion.

VII. Appendix

A. The Explicit Forms Of Linear Governing Equations

1) Strain-displacement Equations: The strain-displacement equations in Eq. (4) can be rewritten explicitly as

$$\begin{aligned}
\varepsilon^s_{xx} &= \frac{\partial d^x_s}{\partial x}, \varepsilon^s_{yy} = \frac{\partial d^y_s}{\partial y}, \varepsilon^s_{zz} = \frac{\partial d^z_s}{\partial z}, \\
\varepsilon^s_{xy} &= \frac{1}{2}(\frac{\partial d^x_s}{\partial y} + \frac{\partial d^y_s}{\partial x}), \\
\varepsilon^s_{yz} &= \frac{1}{2}(\frac{\partial d^y_s}{\partial z} + \frac{\partial d^z_s}{\partial y}), \\
\varepsilon^s_{xz} &= \frac{1}{2}(\frac{\partial d^x_s}{\partial z} + \frac{\partial d^z_s}{\partial x}).
\end{aligned} \tag{30}$$

2) Static Equilibrium Equations: The static equilibrium equation in Eq. (5) can be written in Einstein notation as

$$\sigma^s_{ji,j} + F_i = 0, \tag{31}$$

where $F_i \in \mathbb{R}$ is the body force that is neglected and set as zero at the static equilibrium, $i \in \{x, y, z\}$ and $j \in \{x, y, z\}$ denote three spatial directions. By neglecting the influence of gravity, Eq. (31) can be rewritten explicitly as

$$\begin{aligned}
\frac{\partial \sigma^s_{xx}}{\partial x} + \frac{\partial \sigma^s_{yx}}{\partial y} + \frac{\partial \sigma^s_{zx}}{\partial z} &= 0, \\
\frac{\partial \sigma^s_{xy}}{\partial x} + \frac{\partial \sigma^s_{yy}}{\partial y} + \frac{\partial \sigma^s_{zy}}{\partial z} &= 0, \\
\frac{\partial \sigma^s_{xz}}{\partial x} + \frac{\partial \sigma^s_{yz}}{\partial y} + \frac{\partial \sigma^s_{zz}}{\partial z} &= 0.
\end{aligned} \tag{32}$$

The explicit forms of Eq. (16) are

$$\varepsilon^s_{ii} = \frac{\partial d^i_s}{\partial i} + \frac{1}{2}[(\frac{\partial d^x_s}{\partial i})^2 + (\frac{\partial d^y_s}{\partial i})^2 + (\frac{\partial d^z_s}{\partial i})^2], \tag{33}$$

where $i \in \{x, y, z\}$,

$$\varepsilon^s_{ij} = \frac{1}{2}(\frac{\partial d^i_s}{\partial j} + \frac{\partial d^j_s}{\partial i}) + \frac{1}{2}\Big[\frac{\partial d^x_s}{\partial i}\frac{\partial d^x_s}{\partial j} + \frac{\partial d^y_s}{\partial i}\frac{\partial d^y_s}{\partial j} + \frac{\partial d^z_s}{\partial i}\frac{\partial d^z_s}{\partial j}\Big], \tag{34}$$

with the pair of dual indexes $\langle i, j\rangle \in \{\langle x, y\rangle, \langle x, z\rangle, \langle y, z\rangle\}$.

3) Constitutive Equations: The constitutive equation in Eq. (6) in Einstein notation is

$$\boldsymbol{\sigma}^s_{ij} = \mathsf{C}^{ijkl}_s \boldsymbol{\varepsilon}^s_{kl}. \tag{35}$$

By reloading the notations $\boldsymbol{\sigma}^s$, C_s and $\boldsymbol{\varepsilon}^s$ in Eq. (6) from tensor to vector/matrix forms, i.e., $\boldsymbol{\sigma}^s \in \mathbb{R}^6$, $\mathsf{C}_s \in \mathbb{R}^{6\times 6}$ and $\boldsymbol{\varepsilon}^s \in \mathbb{R}^6$, more specifically as

$$\boldsymbol{\sigma}^s = \left[\sigma^s_{xx}, \sigma^s_{yy}, \sigma^s_{zz}, \sigma^s_{xy}, \sigma^s_{xz}, \sigma^s_{yz}\right]^{\mathsf{T}}, \tag{36}$$

$$\mathsf{C}_s = \begin{bmatrix}
(\lambda_s + 2\mu_s) & \lambda_s & \lambda_s & 0 & 0 & 0 \\
\lambda_s & (\lambda_s + 2\mu_s) & \lambda_s & 0 & 0 & 0 \\
\lambda_s & \lambda_s & (\lambda_s + 2\mu_s) & 0 & 0 & 0 \\
0 & 0 & 0 & \mu_s & 0 & 0 \\
0 & 0 & 0 & 0 & \mu_s & 0 \\
0 & 0 & 0 & 0 & 0 & \mu_s
\end{bmatrix}, \tag{37}$$

$$\boldsymbol{\varepsilon}^s = \left[\varepsilon^s_{xx}, \varepsilon^s_{yy}, \varepsilon^s_{zz}, 2\varepsilon^s_{xy}, 2\varepsilon^s_{xz}, 2\varepsilon^s_{yz}\right]^{\mathsf{T}}, \tag{38}$$

the constitutive equation in Eq. (6) can be rewritten as

$$\boldsymbol{\sigma}^s = \mathsf{C}_s \boldsymbol{\varepsilon}^s. \tag{39}$$

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In Eq. (36), $\lambda_s > 0$ and $\mu_s > 0$ are Lamé parameters, which are computed as

$$\lambda_s = \frac{E_s \nu_s}{(1-2\nu_s)(1+\nu_s)}, \tag{40}$$

and

$$\mu_s = \frac{E_s}{2(1+\nu_s)}, \tag{41}$$

where $E_s > 0$ and $v_s \in (0, 0.5)$ denote the Young's Modulus and Possion's ratio with the s-th source point $\mathbf{p}_s$, respectively.

B. The Explicit Forms of Loss Functions with Linear Elasticity

1) The Loss Term with the Static Equilibrium Equation: The explicit form of $f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s)$, which is used to compute $\mathcal{L}_S^k(\theta_h; \mathcal{P}_k)$ that composes Eq. (14) and Eq. (21), is the sum of squared L2 norm of the left sides of Eq. (32)

$$\begin{aligned} f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) = &\left|\left|\frac{\partial \sigma_{xx}^s}{\partial x} + \frac{\partial \sigma_{yx}^s}{\partial y} + \frac{\partial \sigma_{zx}^s}{\partial z}\right|\right|_2^2 \\ + &\left|\left|\frac{\partial \sigma_{xy}^s}{\partial x} + \frac{\partial \sigma_{yy}^s}{\partial y} + \frac{\partial \sigma_{zy}^s}{\partial z}\right|\right|_2^2 + \left|\left|\frac{\partial \sigma_{xz}^s}{\partial x} + \frac{\partial \sigma_{yz}^s}{\partial y} + \frac{\partial \sigma_{zz}^s}{\partial z}\right|\right|_2^2. \end{aligned} \tag{42}$$

2) The Loss Term with the Constitutive Equation: The explicit form of the loss term $f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, \mathsf{C}_s)$, which is used to compute $\mathcal{L}_C^k(\theta; \mathcal{P}_k)$ that composes Eq. (14) and Eq. (21), is defined as the sum of squared L2 norm of the individual differences between the left and right sides of Eq. (39)

$$\begin{aligned} &f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, \mathsf{C}_s) = f_{\text{const}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s, \lambda_s, \mu_s) = \\ &\big(||(\lambda_s + 2\mu_s)\varepsilon_{xx}^s + \lambda_s(\varepsilon_{yy}^s + \varepsilon_{zz}^s) - \sigma_{xx}^s||_2^2 \\ &+ ||\lambda_s(\varepsilon_{xx}^s + \varepsilon_{zz}^s) + (\lambda_s + 2\mu_s)\varepsilon_{yy}^s - \sigma_{yy}^s||_2^2 \\ &+ ||\lambda_s\varepsilon_{xx}^s + \lambda_s\varepsilon_{yy}^s + (\lambda_s + 2\mu_s)\varepsilon_{zz}^s - \sigma_{zz}^s||_2^2 \\ &+ ||\sigma_{xy}^s - 2\mu_s\varepsilon_{xy}^s||_2^2 + ||\sigma_{xz}^s - 2\mu_s\varepsilon_{xz}^s||_2^2 + ||\sigma_{yz}^s - 2\mu_s\varepsilon_{yz}^s||_2^2\big), \end{aligned} \tag{43}$$

where the strain tensor ε_{ij}^s at $\mathbf{p}_s$ is computed from network-predicted $\mathbf{d}_s$ with the automatic differentiation technique, according to the kinematic equation in Eq. (4), λ_s and μ_s are Lamé parameters computed with Eq. (40) and Eq. (41) using Young's modulus and Possion's ratio.

3) Elastic Energy: The explicit form of the $f_{\text{energy}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s)$ is defined as

$$\begin{aligned} f_{\text{energy}}(\boldsymbol{\varepsilon}^s, \boldsymbol{\sigma}^s) = \frac{1}{2}\big(&\varepsilon_{xx}^s\sigma_{xx}^s + \varepsilon_{yy}^s\sigma_{yy}^s + \varepsilon_{zz}^s\sigma_{zz}^s \\ &+ 2\varepsilon_{xy}^s\sigma_{xy}^s + 2\varepsilon_{xz}^s\sigma_{xz}^s + 2\varepsilon_{yz}^s\sigma_{yz}^s\big). \end{aligned} \tag{44}$$

C. The Explicit Forms Of Nonlinear Governing Equations

1) Nonlinear Strain-displacement Equations: The explicit forms of the nonlinear strain-displacement equation in Eq. (16) are

$$\varepsilon_{ii}^s = \frac{\partial d_s^i}{\partial i} + \frac{1}{2}[(\frac{\partial d_s^x}{\partial i})^2 + (\frac{\partial d_s^y}{\partial i})^2 + (\frac{\partial d_s^z}{\partial i})^2], \tag{45}$$

where $i \in \{x, y, z\}$, and

$$\varepsilon_{ij}^s = \frac{1}{2}(\frac{\partial d_s^i}{\partial j} + \frac{\partial d_s^j}{\partial i}) + \frac{1}{2}\Big[\frac{\partial d_s^x}{\partial i}\frac{\partial d_s^x}{\partial j} + \frac{\partial d_s^y}{\partial i}\frac{\partial d_s^y}{\partial j} + \frac{\partial d_s^z}{\partial i}\frac{\partial d_s^z}{\partial j}\Big], \tag{46}$$

where $\langle i, j\rangle \in \{\langle x, y\rangle, \langle x, z\rangle, \langle y, z\rangle\}$ denotes the pair of dual indexes.

2) Static Equilibrium Equation: The PDEs that compose $\mathcal{L}_S^k(\theta_h; \mathcal{P}_k)$ in Eq. (14) and Eq. (21) are

$$\begin{aligned} f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) = &||\frac{\partial \sigma_{xx}^s}{\partial x} + \frac{\partial \sigma_{yx}^s}{\partial y} + \frac{\partial \sigma_{zx}^s}{\partial z}||_2^2 \\ + &||\frac{\partial \sigma_{xy}^s}{\partial x} + \frac{\partial \sigma_{yy}^s}{\partial y} + \frac{\partial \sigma_{zy}^s}{\partial z}||_2^2 + ||\frac{\partial \sigma_{xz}^s}{\partial x} + \frac{\partial \sigma_{yz}^s}{\partial y} + \frac{\partial \sigma_{zz}^s}{\partial z}||_2^2, \end{aligned} \tag{47}$$

where here $\boldsymbol{\sigma}^s$ is the second Piola-Kirchhoff stress tensor and the derivatives are taken in the reference configuration. If we utilise the Cauchy stress σ_{xx}^s(here we reload the operator again), $f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s)$ becomes

$$\begin{aligned} f_{\text{static}}(\boldsymbol{\sigma}^s, \mathbf{p}_s) = &||\frac{\partial \sigma_{xx}^s}{\partial (x + d_s^x)} + \frac{\partial \sigma_{yx}^s}{\partial (y + d_s^y)} + \frac{\partial \sigma_{zx}^s}{\partial (z + d_s^z)}||_2^2 \\ + &||\frac{\partial \sigma_{xy}^s}{\partial (x + d_s^x)} + \frac{\partial \sigma_{yy}^s}{\partial (y + d_s^y)} + \frac{\partial \sigma_{zy}^s}{\partial (z + d_s^z)}||_2^2 \\ + &||\frac{\partial \sigma_{xz}^s}{\partial (x + d_s^x)} + \frac{\partial \sigma_{yz}^s}{\partial (y + d_s^y)} + \frac{\partial \sigma_{zz}^s}{\partial (z + d_s^z)}||_2^2, \end{aligned} \tag{48}$$

where here the involved derivatives are taken in the deformed configuration.

3) The Expression of $\mathbf{F}_s\mathbf{F}_s^{\mathsf{T}}$: Given the definition $\mathbf{F}_s = \mathbf{I}_{3\times 3} + \frac{\partial \mathbf{d}_s}{\partial \mathbf{p}_s}$, here we present the expression of $\mathbf{F}_s\mathbf{F}_s^{\mathsf{T}}$ in terms of $\boldsymbol{\varepsilon}^s$ as follows

$$\begin{bmatrix} 1 + 2\varepsilon_{xx}^s & 2\varepsilon_{xy}^s & 2\varepsilon_{xz}^s \\ 2\varepsilon_{xy}^s & 1 + 2\varepsilon_{yy}^s & 2\varepsilon_{yz}^s \\ 2\varepsilon_{xz}^s & 2\varepsilon_{yz}^s & 1 + 2\varepsilon_{zz}^s \end{bmatrix}, \tag{49}$$

where it is clearly noted that $\mathbf{F}_s\mathbf{F}_s^{\mathsf{T}}$ is symmetrical.

D. The Matrix Forms and Voigt Notations of Strain and Stress Tensors

The stress tensor $\boldsymbol{\sigma}^s$ in matrix form is defined as

$$\boldsymbol{\sigma}^s = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}. \tag{50}$$

Considering that $\boldsymbol{\sigma}^s$ is symmetric, it can be simplified to a 6-dimensional vector in Voigt notation

$$[\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{xy}, \sigma_{xz}, \sigma_{yz}]^{\mathsf{T}}. \tag{51}$$

Similarly the strain tensor $\boldsymbol{\varepsilon}^s$ in matrix form is

$$\boldsymbol{\varepsilon}^s = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz}, \end{bmatrix}, \tag{52}$$

which can be simplified to the 6-dimensional vector in Voigt notation

$$[\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, 2\varepsilon_{xy}, 2\varepsilon_{xz}, 2\varepsilon_{yz}]^{\mathsf{T}}. \tag{53}$$

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E. Smooth-PINNs

To more clearly/directly illustrate the advantage of the leveraged biomechanics-related constraints against the smoothness-based regularization, we have formulated and compared a variant of our proposed method where the chamfer loss is retained while the most frequently utilised smoothness term [36] is used. Let's refer to this variant of our method as Smooth-PINNs, where the smoothness regularization term is defined as

$$\mathcal{L}^k_{Smooth}(\theta_h;\mathcal{P}_k)=\frac{1}{|\widetilde{N}_s|}\sum_{s=1}^{|\widetilde{N}_s|}||\nabla\mathbf{d}_s||_2^2, \tag{54}$$

where $|\widetilde{N}_s|\in\mathbb{N}^+$ is the number of utilised source points, the displacement gradient $\nabla\mathbf{d}_s$ is defined as $\nabla\mathbf{d}_s=[\frac{\partial\mathbf{d}_s}{\partial p_s^x},\frac{\partial\mathbf{d}_s}{\partial p_s^y},\frac{\partial\mathbf{d}_s}{\partial p_s^z}]$, whose detailed matrix form is

$$\nabla\mathbf{d}_s=\begin{bmatrix}\frac{\partial d_s^x}{\partial p_s^x} & \frac{\partial d_s^x}{\partial p_s^y} & \frac{\partial d_s^x}{\partial p_s^z}\\ \frac{\partial d_s^y}{\partial p_s^x} & \frac{\partial d_s^y}{\partial p_s^y} & \frac{\partial d_s^y}{\partial p_s^z}\\ \frac{\partial d_s^z}{\partial p_s^x} & \frac{\partial d_s^z}{\partial p_s^y} & \frac{\partial d_s^z}{\partial p_s^z}\end{bmatrix}. \tag{55}$$

The detailed expression of $\mathcal{L}^k_{Smooth}(\theta_h;\mathcal{P}_k)$ is

$$\begin{aligned}\mathcal{L}^k_{Smooth}(\theta_h;\mathcal{P}_k)=\frac{1}{|\widetilde{N}_s|}\sum_{s=1}^{|\widetilde{N}_s|}\Big((\frac{\partial d_s^x}{\partial p_s^x})^2+(\frac{\partial d_s^y}{\partial p_s^x})^2+(\frac{\partial d_s^z}{\partial p_s^x})^2\\ +(\frac{\partial d_s^x}{\partial p_s^y})^2+(\frac{\partial d_s^y}{\partial p_s^y})^2+(\frac{\partial d_s^z}{\partial p_s^y})^2\\ +(\frac{\partial d_s^x}{\partial p_s^z})^2+(\frac{\partial d_s^y}{\partial p_s^z})^2+(\frac{\partial d_s^z}{\partial p_s^z})^2\Big).\end{aligned} \tag{56}$$

Then the overall training loss for the smoothness-regularized variant of our proposed approach is defined as

$$w\mathcal{L}^k_R(\theta_g;\mathcal{P}_k)+\mathcal{L}^k_{Smooth}(\theta_h;\mathcal{P}_k), \tag{57}$$

where $\mathcal{L}^k_R(\theta_g;\mathcal{P}_k)\in\mathbb{R}$ is the Chamfer loss defined in Eq. (7), $w\in\mathbb{R}$ representing the corresponding weight.

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